University of Haripur

Khyber Pakhtoonkhwa

Lab Reports

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Lab Number #01

What is UNIX?

- The UNIX is operating system capable of handling activities from multiple users
- The UNIX operating system is a set of programs that acts as a link between the computer and user.
- The computer programs that allocate system resources and coordinate all the details of the computer internals is called the operating system or kernel
- Users communicate with kernel through a program known as shell

UNIX Artitecture?

Kernel: The kernel is the heart of the operating system. It interacts with hardware and most of the tasks like memory management, task scheduling and file management

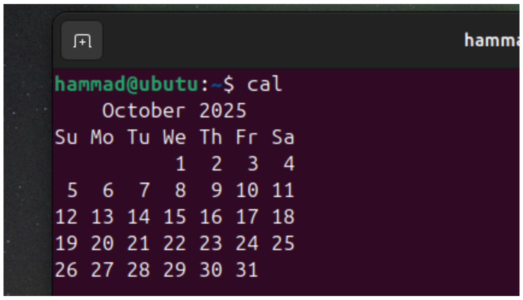
Shell: The shell is the utility that processes your requests. When you type in a command at your terminal, the shell interprets the command and calls the program that you want. The shell uses standard syntax for all commands. C Shell, Bourne Shell and Korn Shell are most famous shells which are available with most of the Unix variants.

Login

- When you first connect to a UNIX system, you usually see a prompt such as the following:
  login:
- Have your userid (user identification) and password ready.
- Type your userid at the login prompt, then press ENTER. Your userid is case-sensitive
- Type your password at the password prompt, then press ENTER. Your password is also case-sensitive.
- If you provided correct userid and password then you would be allowed to enter into the system.

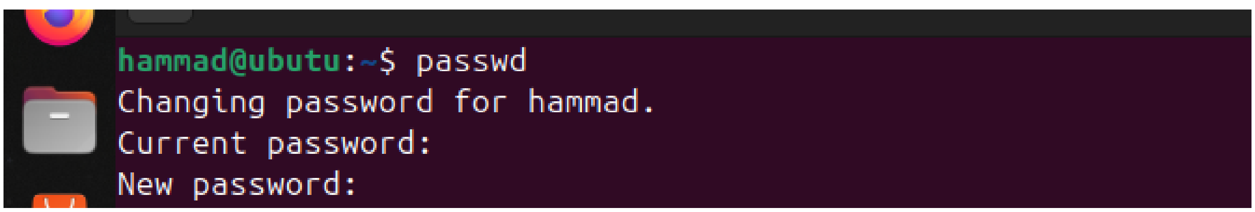
To view Calender

You would be provided with a command prompt (sometime called $ prompt) where you would type your all the commands. For example to check calendar you need to type cal command as follows:



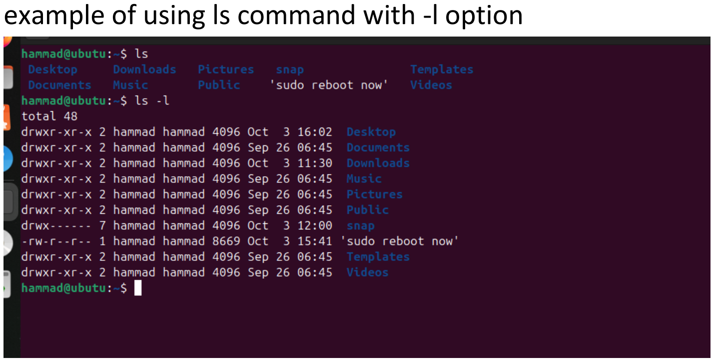
Change Password

All Unix systems require passwords to help ensure that your files and data remain your own and that the system itself is secure from hackers and crackers. Here are the steps to change your password:
To start, type passwd at command prompt as shown below.
Enter your old password the one you're currently using.
Type in your new password. Always keep your password complex enough so that no body can guess it. But make sure, you remember it.
You would need to verify the password by typing it again.



Listing Directories and Files

All data in UNIX is organized into files. All files are organized into directories. These directories are organized into a tree-like structure called the filesystem.
You can use ls command to list out all the files or directories available in a directory. Following is the example of using ls command with -l option



Directory Commands

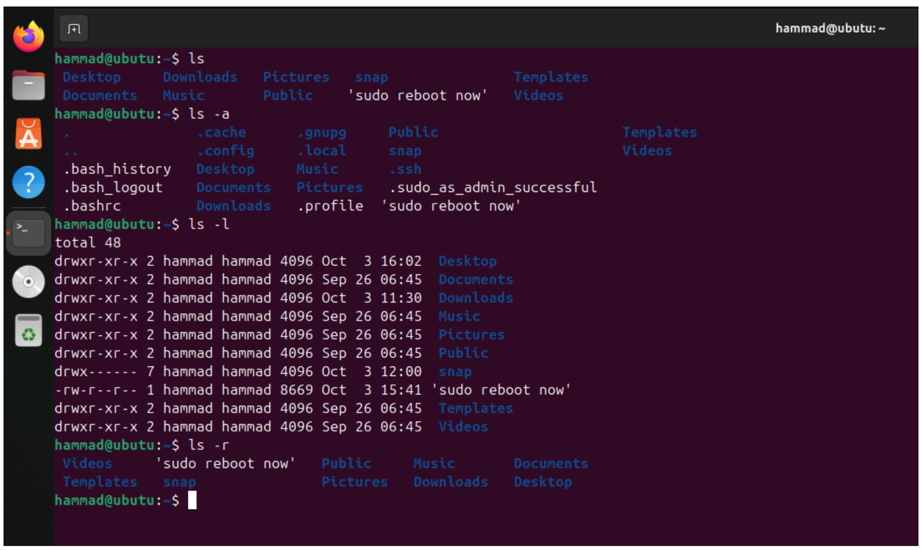
Command	Description
ls	List the file in the directory, just like ***dir*** command in DOS.

Option:

-a Display all the files, and subdirectories, including hidden files.

-l Display detailed information about each file, and directory.

-r - Display files in the reverse order.



Who are you?

While you're logged in to the system, you might be willing to know : Who am I?
The easiest way to find out "who you are" is to enter the whoami command:

$ whoami

Who is Logged in?

- Sometime you might be interested to know who is logged in to the computer at the same time
- There are three commands are available to get you this information, based on how much you'd like to learn about the other users: users, who, and w.
  $ users
  $ who
- Try w command on your system to check the output. This would list down few more information associated with the users logged in the system.

```
hammad@ubutu:~$ who
hammad   seat0        2025-10-11 01:10
hammad   tty2         2025-10-11 01:10
hammad@ubutu:~$ user
Command 'user' not found, did you mean:
  command 'fuser' from deb psmisc (23.7-2)
  command 'users' from deb coreutils (9.5-1ubuntu1.25.04.2)
Try: sudo apt install <deb name>
hammad@ubutu:~$ users
hammad hammad
hammad@ubutu:~$ w
 01:27:07 up 17 min,  2 users,  load average: 0.02, 0.06, 0.11
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU  WHAT
hammad   tty2     -                01:10   17:30   0.06s  0.06s /usr/libexec/gnome-session-binary --session=ubuntu
hammad            -                01:10   17:15   0.00s  1.66s /usr/lib/systemd/systemd --user
hammad@ubutu:~$
```

How to Shutdown the system?

- The most consistent way to shut down a Unix system properly via the command line is to use one of the following commands:
  $poweroff
  $reboot
  $shutdown
- You typically need to be the superuser or root (the most privileged account on a Unix system) to shut down the system

Create, Display and Merge Command

Command: cat —cat" is short for concatenate. This command is used to create, view and concatenate files.

Example:

- cat>file1

This command creates the file, enter the contents into it and to save the content, press Ctrl+D.

- cat file1

This command shows the content of the file.

- cat file1 file2 > file3

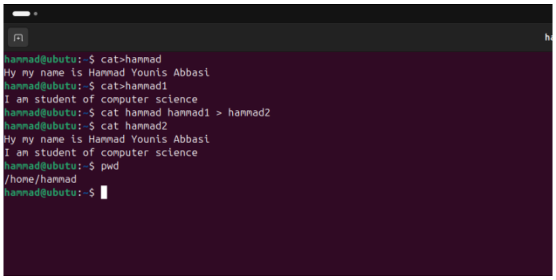
This command combines the contents of the first two files into the third file.

Command: pwd "pwd" stands for print working directory. It displays your current position in the UNIX/LINUX file system.

Example:

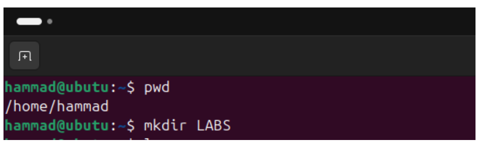
- pwd

It is simply used to report your current working directory.

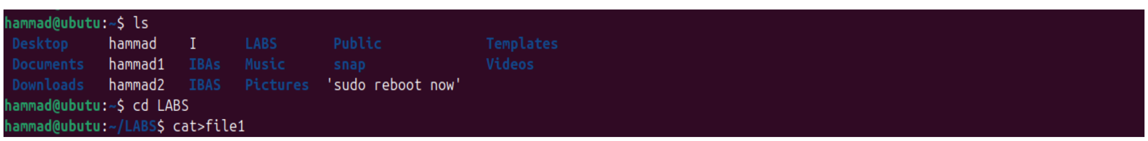


Activities

Activity #01: Verify that you are in your home directory. Make the directory LABS using the following comman



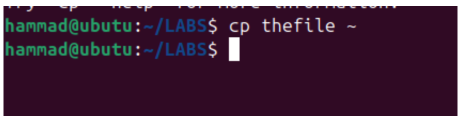
Activity #02: List the files in the current directory to verify that the directory LABS has been made correctly Change directories to LABS



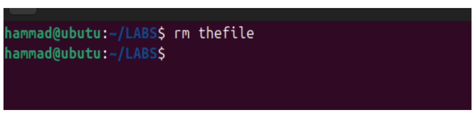
Activity #03: Create the file named file? List the contents of the file file1 to the screen. Make a copy of the file filed under the name file2. Verify that the files file 1 and file2 both exist.

Activity #04: List the contents of both file and file2 to the monitor screen. Then delete the file file1. Clear the window. Rename file2 to thefile

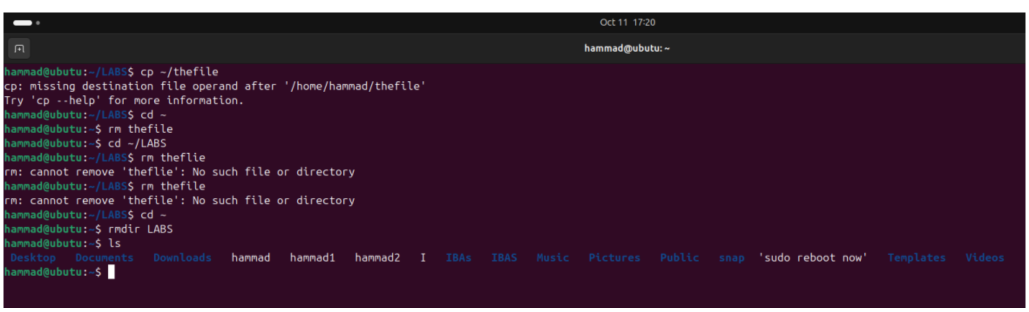
Activity #05: Copy thefile to your home directory.



Activity #06: Remove thefile from the current directory



Activity #07: Copy thefile from your home directory to the current directory Change directories to your home directory Remove thefile from your home directory and from directory LABS. Verify thefile is removed from the directory LABS Remove the directory LABS from your home directory with the following command. Verify that thefile and LABS are gone from your home directory



Lab number #02

Files: All data in UNIX is organized into files. All files are organized into directories. These directories are organized into a tree-like structure called the filesystem.

When you work with UNIX, one way or another you spend most of your time working with files. This lecture would teach you how to create and remove files, copy and rename them, create links to them etc.

Types of Files:

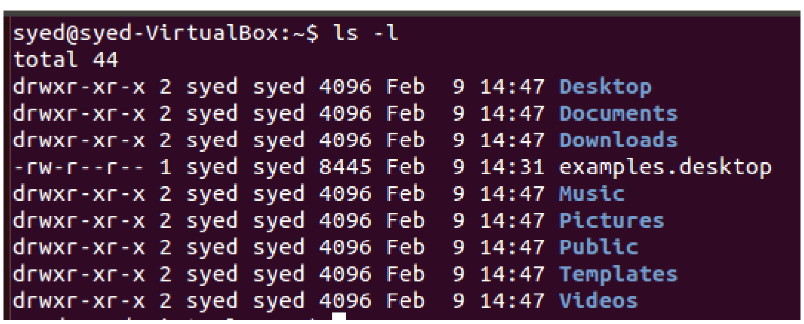
In UNIX there are three basic types of files:

1. Ordinary Files: An ordinary file is a file on the system that contains data, text, or program instructions. In this tutorial, you look at working with ordinary files.

2. Directories: Directories store both special and ordinary files. For users familiar with Windows or Mac OS, UNIX directories are equivalent to folders.

3. Special Files: Some special files provide access to hardware such as hard drives, CD-ROM drives, modems, and Ethernet adapters. Other special files are similar to aliases or shortcuts and enable you to access a single file using different names

Ls –l Command: The command ls supports the -l option which would help you to get more information about the listed files:



```
syed@syed-VirtualBox:~$ ls -l
total 44
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Desktop
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Documents
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Downloads
-rw-r--r-- 1 syed syed 8445 Feb  9 14:31 examples.desktop
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Music
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Pictures
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Public
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Templates
drwxr-xr-x 2 syed syed 4096 Feb  9 14:47 Videos
```

Here is the information about all the listed columns:

1. First Column: represents file type and permission given on the file. Below is the description of all type of files.
2. Second Column: represents the number of memory blocks taken by the file or directory.
3. Third Column: represents owner of the file. This is the Unix user who created this file.
4. Fourth Column: represents group of the owner. Every Unix user would have an associated group.
5. Fifth Column: represents file size in bytes.
6. Sixth Column: represents date and time when this file was created or modified last time.
7. Seventh Column: represents file or directory name.

Meta Character: Meta characters have special meaning in Unix. For example * and ? are metacharacters. We use * to match 0 or more characters, a question mark ? matches with single character.

$ls ch*.doc

Displays all the files whose name start with ch and ends with .doc:

ch01-1.doc ch010.doc ch02.doc ch03-2.doc

ch04-1.doc ch040.doc ch05.doc ch06-2.doc

ch01-2.doc ch02-1.doc

Here * works as meta character which matches with any character. If you want to display all the files ending with just .doc then you can use following command:

$ls *.doc

Hidden File:

An invisible file is one whose first character is the dot (.). UNIX programs (including the shell) use most of these files to store configuration information.

Some common examples of hidden files include the files:

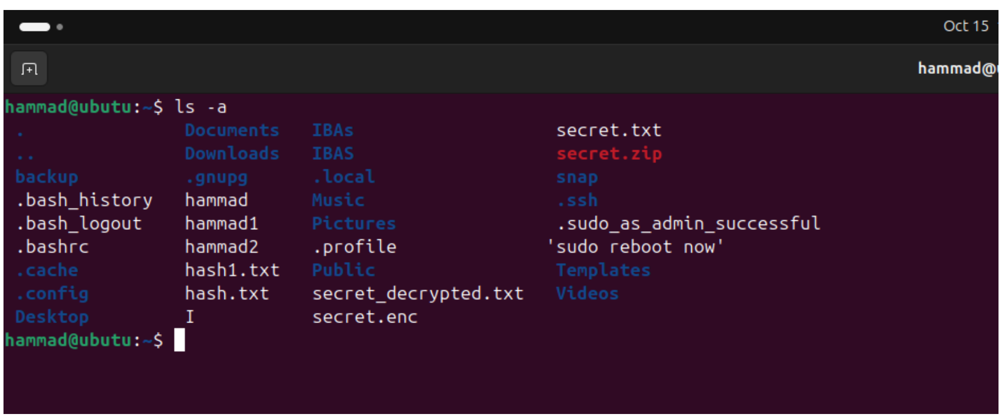
.profile: the Bourne shell (sh) initialization script

.kshrc: the Korn shell (ksh) initialization script

.cshrc: the C shell (csh) initialization script

.rhosts: the remote shell configuration file

To list invisible files, specify the -a option to ls:



Vim Editor

sudo apt-get install vim

Command mode: letters or sequence of letters interactively command vim. Commands are case sensitive. The ESC key can end a command.

Edit mode: Text is inserted. The ESC key ends Edit mode and returns you to command mode. One can enter edit mode with the "i"

:q

quitting vim editor

:wq filename
saving and quitting

Creating File: You can use vi editor to create ordinary files on any Unix system. You simply need to give following command:

$ vi filename

Above command would open a file with the given filename. You would need to press key i to come into edit mode.

Once you are done, do the following steps:
Press key esc to come out of edit mode.
Press two keys Shift + ZZ together to come out of the file completely.

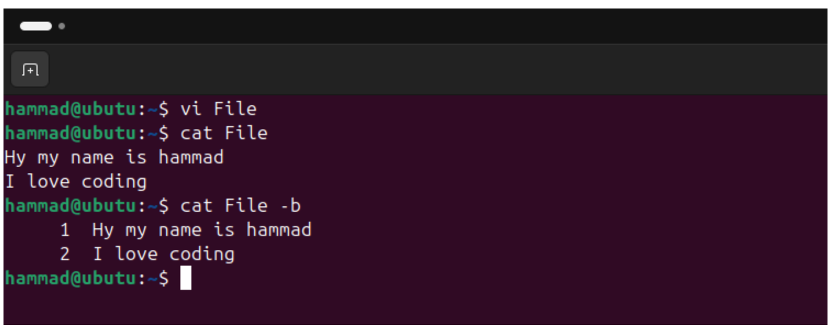
Display The Content of a File:

You can use cat command to see the content of a file. Following is the simple example to see the content of above created file:

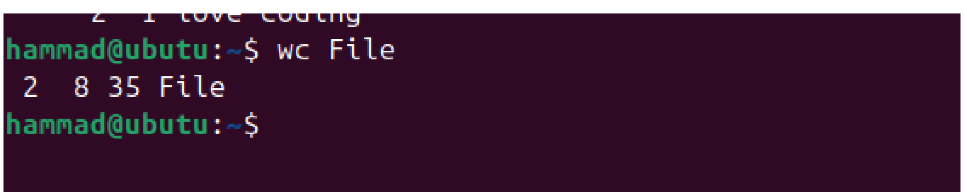
$ cat filename

You can display line numbers by using -b option along with cat command as follows:

$ cat filename -b



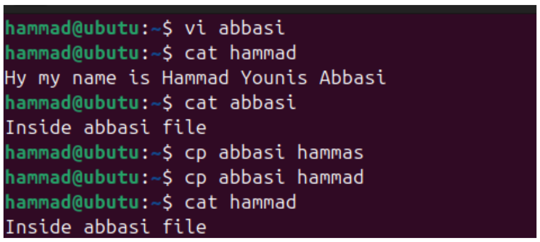
Counting Words in a File: You can use the wc command to get a count of the total number of lines, words, and characters contained in a file. Following is the simple example to see the information about above created file:
$ wc filename



Here is the detail of all the four columns:
First Column: represents total number of lines in the file.
Second Column: represents total number of words in the file.
Third Column: represents total number of bytes in the file. This is actual size of the file.
Fourth Column: represents file name.

Copying File: To make a copy of a file use the cp command. The basic syntax of the command is:

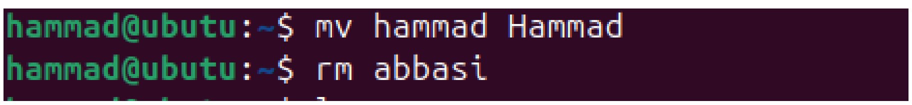
$ cp source_file destination_file



```
hammad@ubutu:~$ vi abbasi
hammad@ubutu:~$ cat hammad
Hy my name is Hammad Younis Abbasi
hammad@ubutu:~$ cat abbasi
Inside abbasi file
hammad@ubutu:~$ cp abbasi hammas
hammad@ubutu:~$ cp abbasi hammad
hammad@ubutu:~$ cat hammad
Inside abbasi file
```

Renaming File: To change the name of a file use the mv command. Its basic syntax is:

$ mv old_file new_file

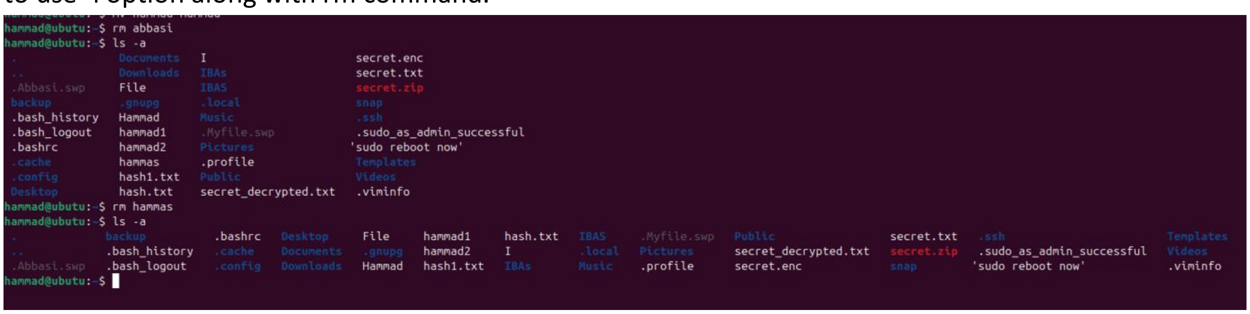


```
hammad@ubutu:~$ mv hammad Hammad
hammad@ubutu:~$ rm abbasi
```

Removing File: To delete an existing file use the rm command. Its basic syntax is:

$ rm filename

Deleting a file can delete useful information. So be careful while using this command. It is recommended to use -i option along with rm command.



```
hammad@ubutu:~$ rm abbasi
hammad@ubutu:~$ ls -a
.              Documents  I                      secret.enc
..             Downloads  IBAs                   secret.txt
.Abbasi.swp    File       IBAS                   secret.zip
backup         .gnupg     .local                 snap
.bash_history  Hammad     Music                  .ssh
.bash_logout   hammad1    .Myfile.swp            .sudo_as_admin_successful
.bashrc        hammad2    Pictures              'sudo reboot now'
.cache         hammas     .profile               Templates
.config        hash1.txt  Public                 Videos
Desktop        hash.txt   secret_decrypted.txt   .viminfo
hammad@ubutu:~$ rm hammas
hammad@ubutu:~$ ls -a
.            backup         .bashrc  Desktop    File    hammad1    hash.txt  IBAS    .Myfile.swp  Public                secret.txt  .ssh                       Templates
..           .bash_history  .cache   Documents  .gnupg  hammad2    I         .local  Pictures     secret_decrypted.txt  secret.zip  .sudo_as_admin_successful  Videos
.Abbasi.swp  .bash_logout   .config  Downloads  Hammad  hash1.txt  IBAs      Music   .profile     secret.enc            snap       'sudo reboot now'           .viminfo
hammad@ubutu:~$
```

Standard UNIX Streams:

Under normal circumstances every Unix program has three streams (files) opened for it when it starts up:

stdin : This is referred to as standard input and associated file descriptor is 0. This is also represented as STDIN. Unix program would read default input from STDIN.

stdout : This is referred to as standard output and associated file descriptor is 1. This is also represented as STDOUT. Unix program would write default output at STDOUT stderr : This is referred to as standard error and associated file descriptor is 2. This is also represented as STDERR. Unix program would write all the error message at STDERR.

Lab Number #03

UNIX Directories: A directory is a file whose sole job is to store file names and related information. All files whether ordinary, special, or directory, are contained in directories.
UNIX uses a hierarchical structure for organizing files and directories. This structure is often referred to as a directory tree . The tree has a single root node, the slash character (/), and all other directories are contained below it.

Home Directories: The directory in which you find yourself when you first login is called your home directory.
You will be doing much of your work in your home directory and subdirectories that you'll be creating to organize your files.
You can go in your home directory anytime using the following command:

$cd ~

Absolute/Relative Pathnames: Directories are arranged in a hierarchy with root (/) at the top. The position of any file within the hierarchy is described by its pathname.
Elements of a pathname are separated by a /. A pathname is absolute if it is described in relation to root, so absolute pathnames always begin with a /.
These are some example of absolute filenames
/etc/passwd

Listing Directories: To list the files in a directory you can use the following syntax:

$ls dirname

```
hammad@ubutu:~$ ls abbasi
hammad@ubutu:~$ cat > abbasi
bash: abbasi: Is a directory
```

Creating Directories: Directories are created by the following command:

$mkdir dirname

Here, directory is the absolute or relative pathname of the directory you want to create. For example, the command:

$mkdir mydir

Creates the directory mydir in the current directory.
Here is another example:

$mkdir /tmp/test-dir

This command creates the directory test-dir in the /tmp directory. The mkdir command produces no output if it successfully creates the requested directory.

```
hammad@ubutu:~$ mkdir Abbasi
mkdir: cannot create directory ‘Abbasi’: File exists
hammad@ubutu:~$ mkdir abbasi
hammad@ubutu:~$ ls abbasi
```

Removing Directories: Directories can be deleted using the rmdir command as follows:

$rmdir dirname

Note: To remove a directory make sure it is empty which means there should not be any file or sub-directory inside this directory.

You can remove multiple directories at a time as follows:

$rmdir dirname1 dirname2 dirname3

Above command removes the directories dirname1, dirname2, and dirname2 if they are empty. The rmdir command produces no output if it is successful.

```
hammad@ubutu:~$ rmdir abbasi
hammad@ubutu:~$ ls abbasi
ls: cannot access 'abbasi': No such file or directory
hammad@ubutu:~$ mkdir file1
```

Changing Directories: You can use the cd command to do more than change to a home directory: You can use it to change to any directory by specifying a valid absolute or relative path. The syntax is as follows:

$cd dirname

Renaming Directories: The mv (move) command can also be used to rename a directory. The syntax is as follows:

$mv olddir newdir

You can rename a directory dir1 to dir2 as follows:

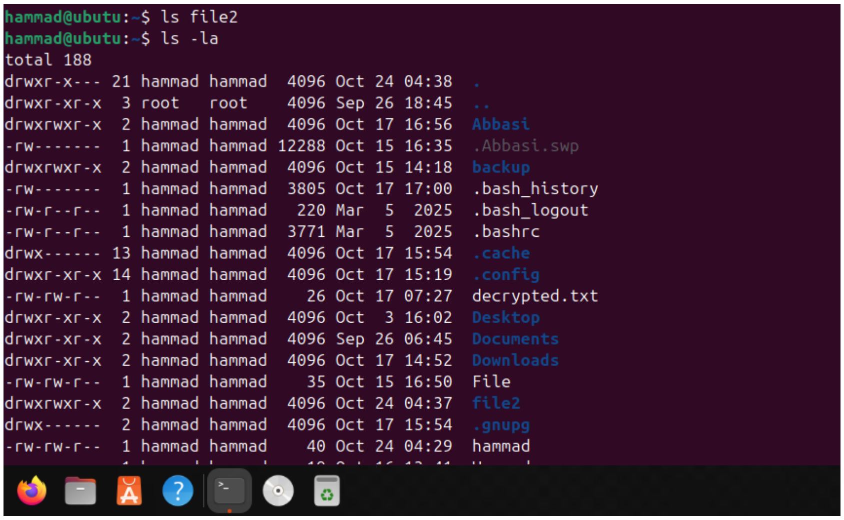
$mv dir1 dir2

```
hammad@ubutu:~$ cd file1
hammad@ubutu:~/file1$ mv file1 file2
mv: cannot stat 'file1': No such file or directory
hammad@ubutu:~/file1$ cd ~
hammad@ubutu:~$ mv file1 file2
hammad@ubutu:~$ ls file2
```

The directories . (dot) and .. (dot dot): The filename . (dot) represents the current working directory; and the filename .. (dot dot) represent the directory one level above the current working directory, often referred to as the parent directory.

If we enter the command to show a listing of the current working directories files and use the -a option to list all the files and the -l option provides the long listing, this is the result.

$ls -la



Lab Number #04:

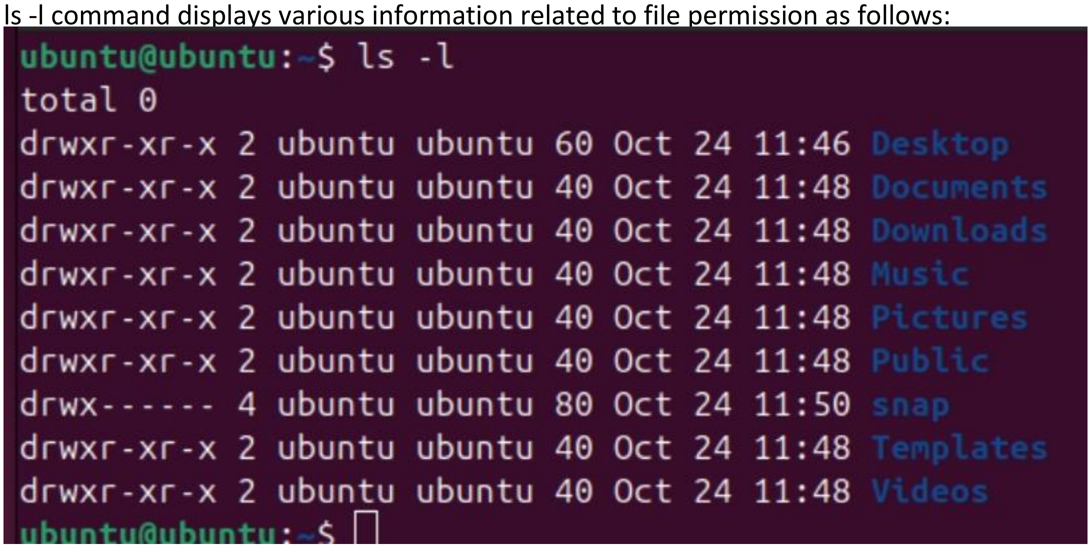
UNIX File Permissions Setup:

File permissions:

File ownership is an important component of UNIX that provides a secure method for storing files. Every file in UNIX has the following attributes: Owner permissions: The owner's permissions determine what actions the owner of the file can perform on the file. Group permissions: The group's permissions determine what actions a user, who is a member of the group that a file belongs to, can perform on the file. Other (world) permissions: The permissions for others indicate what action all other users can perform on the file.

The Permission Indicators:

ls -l command displays various information related to file permission as follows:



The Permission Indicators:

Here first column represents different access mode ie. permission associated with a file or directory. The permissions are broken into groups of threes, and each position in the group denotes a specific permission, in this order: read (r), write (w), execute (x): The first three characters (2-4) represent the permissions for the file's owner. For example –drwxr-xr-x- represents that ower has read (r), write (w) and execute (x) permission. (User) The second group of three characters (5-7) consists of the permissions for the group to which the file belongs. For example -rwxr-xr-- represents that group has read (r) and execute (x) permission but no write permission. (Group)The last group of three characters (8-10) represents the permissions for everyone else. For example -rwxr-xr-- represents that other world has read (r) only permission. (Others)

File Access Mods:

The basic building blocks of Unix permissions are the read, write, and execute permissions, which are described below: Read: Grants the capability to read ie. view the contents of the file. Write: Grants the capability to modify, or remove the content of the file. Execute: User with execute permissions can run a file as a program.

Directory Access Mods:

Directory access modes are listed and organized in the same manner as any other file. There are a few differences that need to be mentioned: Read: Access to a directory means that the user can read the contents. The user can look at the filenames inside the directory. Write: Access means that the user can add or delete files to the contents of the directory. Execute: Executing a directory doesn't really make a lot of sense so think of this as a traverse permission. A user must have execute access to the bin directory in order to execute ls or cd command.

Changing Permissions:

To change file or directory permissions, use the chmod (change mode) command. There are two ways to use chmod: 1)symbolic mode 2)absolute mode

Using Ch Mode in Symbolic Mode:

With symbolic permissions you can add, delete, or specify the permissions using the operators in the following table.

Chmod operator	Description
+	Adds the designated permission(s) to a file or directory.
-	Removes the designated permission(s) from a file or directory.
=	Sets the designated permission(s).

Here's an example using testfile. Running ls -l on testfile shows that the file's permissions are as follows:

```
ubuntu@ubuntu:~$ ls -l testfile
-rw-rw-r-- 1 ubuntu ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Adding write and execute permissions for others permissions to test file$chmod o+wx testfile

```
ubuntu@ubuntu:~$ chmod o+wx testfile
ubuntu@ubuntu:~$ chmod o +wx testfile
chmod: invalid mode: ‘o’
Try 'chmod --help' for more information.
ubuntu@ubuntu:~$ ls -l testfile
-rw-rw-rwx 1 ubuntu ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Removing write permissions for group on testfile $chmod g-w testfile

```
ubuntu@ubuntu:~$ chmod g-w testfile
ubuntu@ubuntu:~$ ls -l testfile
-rw-r--rwx 1 ubuntu ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Removing read permissions for owner (user) on testfile $chmod u-r testfile

```
ubuntu@ubuntu:~$ chmod u-r testfile
ubuntu@ubuntu:~$ ls -l testfile
--w-r--rwx 1 ubuntu ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Using chmod with Absolute Permissions:

The second way to modify permissions with the chmod command is to use a number to specify each set of permissions for the file. Each permission is assigned a value, as the following table shows, and the total of each set of permissions provides a number for that set.

Number	Octal Permission Representation	Ref
0	No permission	---
1	Execute permission	--x
2	Write permission	-w-
3	Execute and write permission: 1 (execute) + 2 (write) = 3	-wx
4	Read permission	r--
5	Read and execute permission: 4 (read) + 1 (execute) = 5	r-x
6	Read and write permission: 4 (read) + 2 (write) = 6	rw-
7	All permissions: 4 (read) + 2 (write) + 1 (execute) = 7	rwx

Here's an example using testfile. Running ls -1 on testfile shows that the file's permissions are as follows:

```
ubuntu@ubuntu:~$ chmod 754 testfile
ubuntu@ubuntu:~$ ls -l testfile
-rwxr-xr-- 1 ubuntu ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Then each example chmod command from the preceding table is run on testfile, followed by ls -l so you can see the permission changes:

```
ubuntu@ubuntu:~$ chmod 444 testfile
ubuntu@ubuntu:~$ ls -l testfile
-r--r--r-- 1 ubuntu ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Changing Owners and Groups:

While creating an account on Unix, it assigns a owner ID and a group ID to each user. All the permissions mentioned above are also assigned based on Owner and Groups. Two commands are available to change the owner and the group of files: chown: The chown command stands for "change owner" and is used to change the owner of a file.

chgrp: The chgrp command stands for "change group" and is used to change the group of a file.

Changing Ownership:

Syntax:$chown username filenameExmaple:$chown user2 testfileChanges the owner of the testfile to the user user2 The super user, root, has the unrestricted capability to change the ownership of a any file but normal users can change only the owner of files they own.

```
ubuntu@ubuntu:~$ sudo chown root testfile
ubuntu@ubuntu:~$ ls -l testfile
-r--r--r-- 1 root ubuntu 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Changing Group Ownership:

Syntax:chgrp groupname filename Example:chgrp group2 testfileChanges the group of the testfile to group2

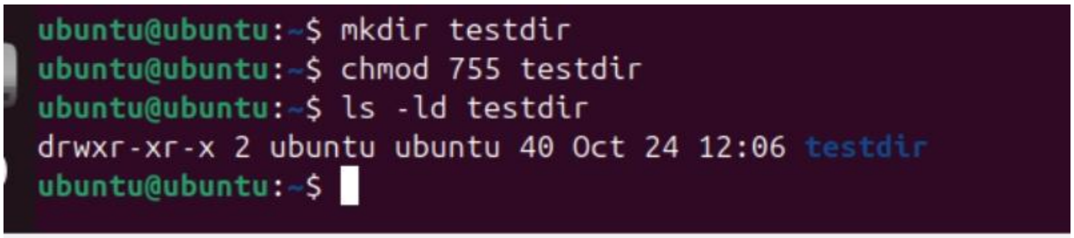
```
ubuntu@ubuntu:~$ sudo chgrp sudo testfile
ubuntu@ubuntu:~$ ls -l testfile
-r--r--r-- 1 root sudo 0 Oct 24 11:55 testfile
ubuntu@ubuntu:~$
```

Create Directory and Practise Directory Permissions:

1)mkdir   testdir

2)chmod 755 testdir

3)ls -ld testdir



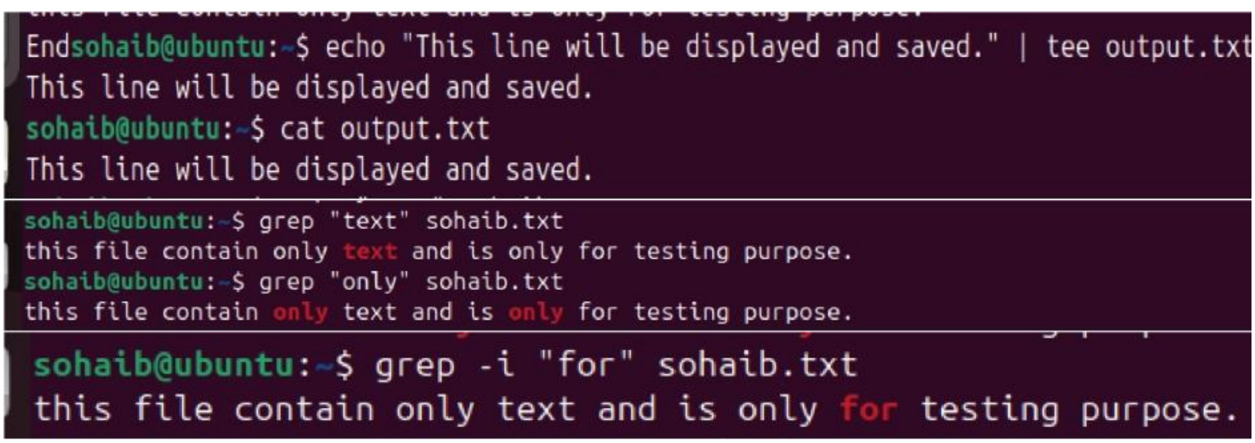
Lab Number #06:

Task 1: Using the tee Command

The tee command reads from standard input and writes both to standard output and to a file.

Command:

echo "This line will be displayed and saved." | tee output.txt



Task 2: Using the grep Command

Search for lines containing a specific string in a file.

Command 2.1 : Basic Search

Command 2.2: Case-Insensitive

Search (-i**)** grep -i "hello" file1.txt

Command 2.3: Invert Match (-v**) - Show lines that do NOT contain the**

pattern grep -v "test" file1.txt

Command 2.4: Show Lines Around

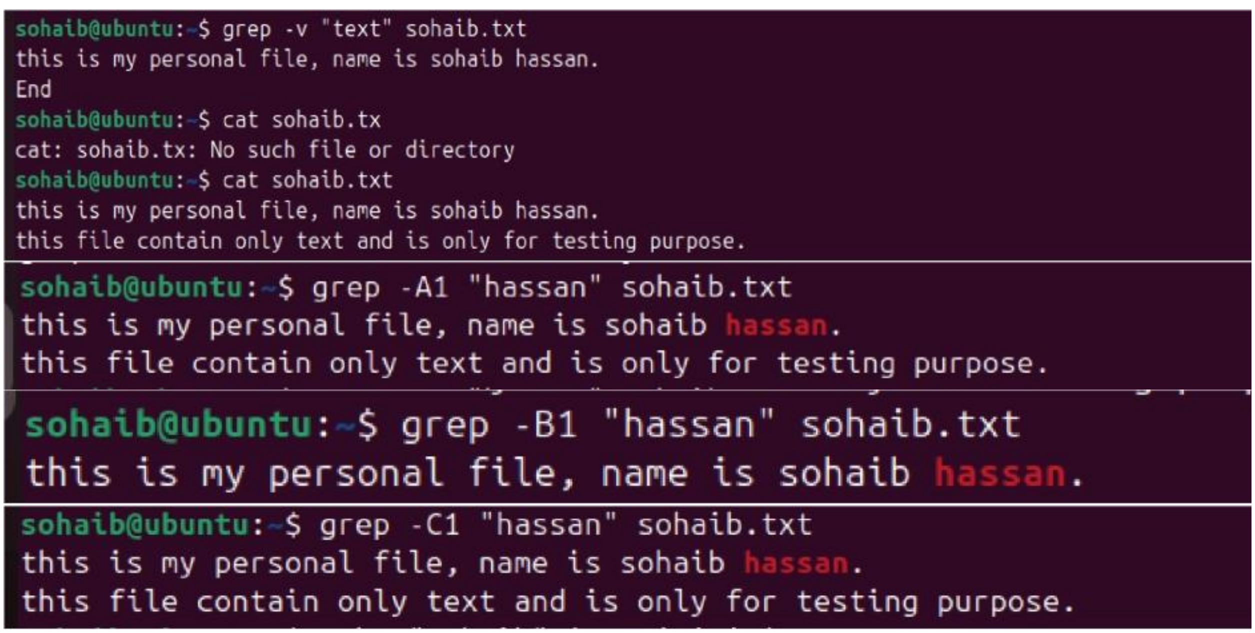
Match (-A**,** -B**,** -C**)** ***Show 1 line after the***

match grep -A1 "multiple" file1.txt

Show 1 line before the

match grep -B1

"world" file1.txt



Show 1 line before and 1 line after the match

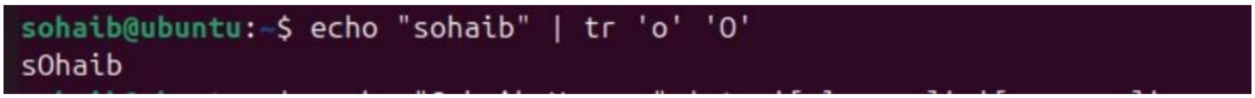
(Context) grep -C1 "hello" file1.txt

Task 3: Using the tr Command

Translate or delete characters.

Command 3.1: Translate one character to another

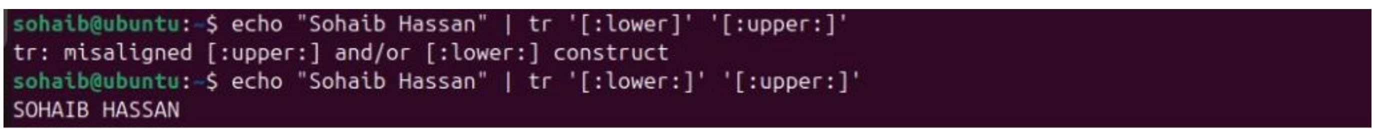
echo "hello" | tr 'e' 'E'



Command 3.2: Convert lowercase to

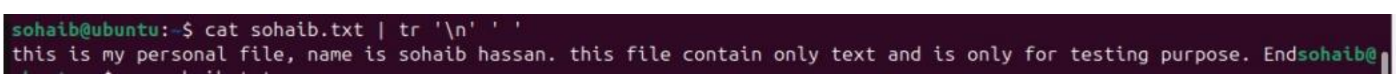
uppercase echo "Hello World" | tr

'[:lower:]' '[:upper:]'



Command 3.3: Translate newlines to spaces (combine lines into one line)

cat file1.txt | tr '\n' ' '



Task 4: Using the wc Command

Count lines, words, and characters in a file.

Command: wc file1.txt

```
ubuntu:~$ wc sohaib.txt
  2  21 113 sohaib.txt
```

Task 5: Using the sort Command

Sort the lines of a text file alphabetically.

Command: sort unsorted.txt

```
sohaib@ubuntu:~$ sort sohaib.txt
End
this file contain only text and is only for testing purpose.
this is my personal file, name is sohaib hassan.
```

Task 6: Using the uniq Command

Remove duplicate lines from a **sorted** file.

uniq only removes *adjacent* duplicates, so the input must be sorted

first. **Command: sort unsorted.txt | uniq**

```
sohaib@ubuntu:~$ sort sohaib.txt | uniq
End
this file contain only text and is only for testing purpose.
this is my personal file, name is sohaib hassan.
```

Task 7: Combining Commands (Pipeline)

This is a powerful concept. Use the output of one command as the input for the next using the pipe | operator.

Command 7.1: Count how many times the word "test" (case-insensitive) appears in file1.txt. grep -i "test" file1.txt | wc -l

```
sohaib@ubuntu:~$ grep -i "hassan" sohaib.txt | wc -1
wc: invalid option -- '1'
Try 'wc --help' for more information.
```

Command 7.2: List all unique words from a file.

cat file1.txt | tr ' ' '\n' | sort | uniq

Lab Number #07:

Commands & Arguments (REDUCED & COMBINED FORMAT)

TOPIC 1: Command Line Scan, Arguments & White Space Handling

Description:

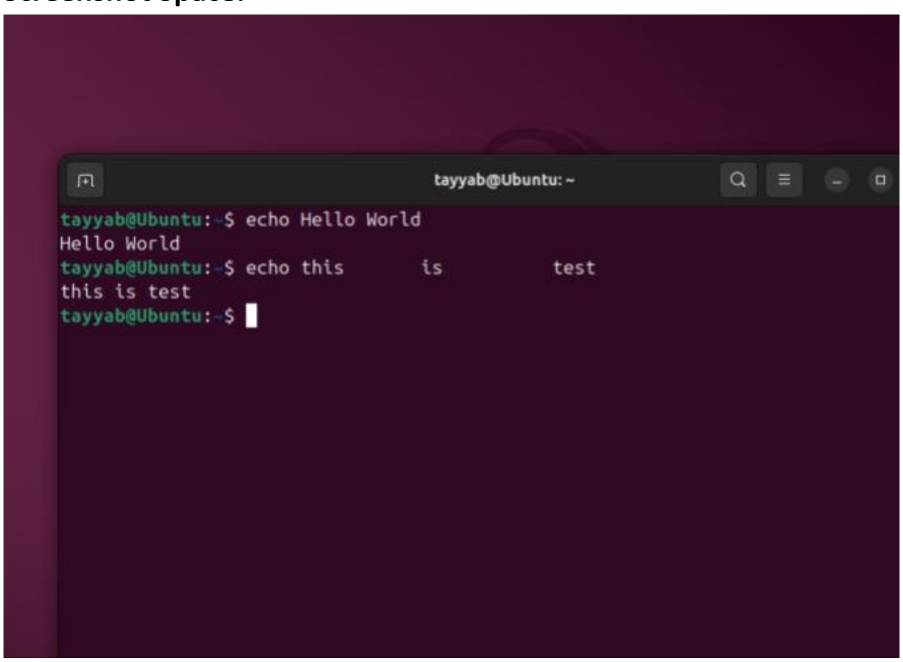
The shell reads a command line, splits it into arguments, and removes extra white spaces. Multiple spaces are treated as one unless quotes are used

Commands:

echo Hello World

echo this   is   test

Screenshot Space:



TOPIC 2: Quoting Mechanisms (Single & Double Quotes)

Description:

Quotes prevent the shell from removing spaces or modifying text.

- Single quotes preserve everything exactly.
- Double quotes preserve spaces but allow variable expansion.

Commands:

echo 'multiple   spaces   kept'

echo "double quotes also keep   spaces"

Screenshot Space:

TOPIC 3: echo with Escape Sequences (echo -e)

Description:

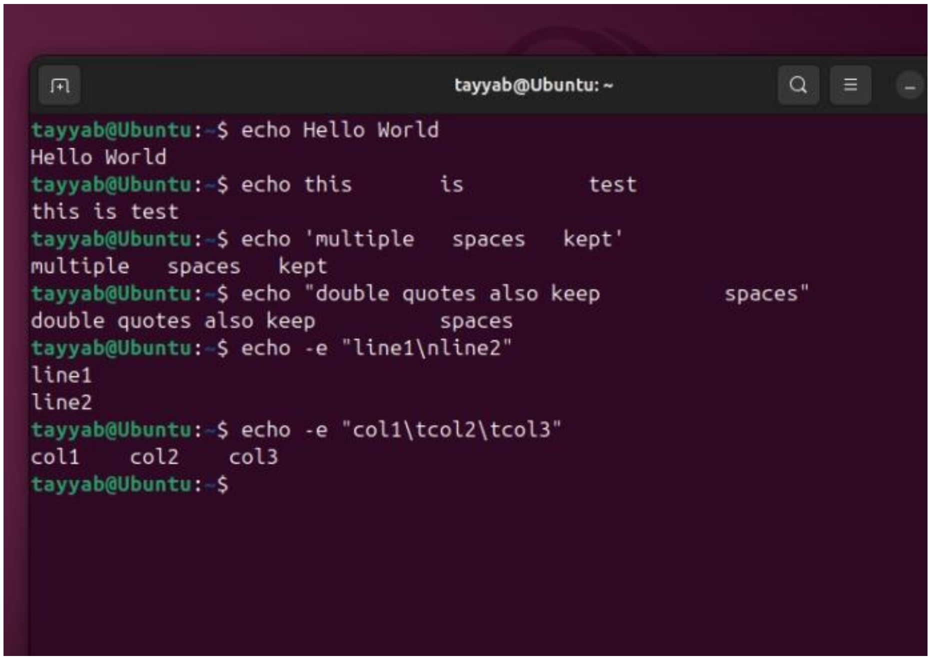
The -e option enables special characters like newline (\n) and tab (\t).

Commands:

echo -e "line1\nline2"

echo -e "col1\tcol2\tcol3"

Screenshot Space:



TOPIC 4: Built-in vs External Commands (type & full path)

Description:

- Some commands are built into the shell (e.g., cd).
- Others are external programs in /bin, /usr/bin.
- type tells which one it is.
- Full path runs the external version explicitly.

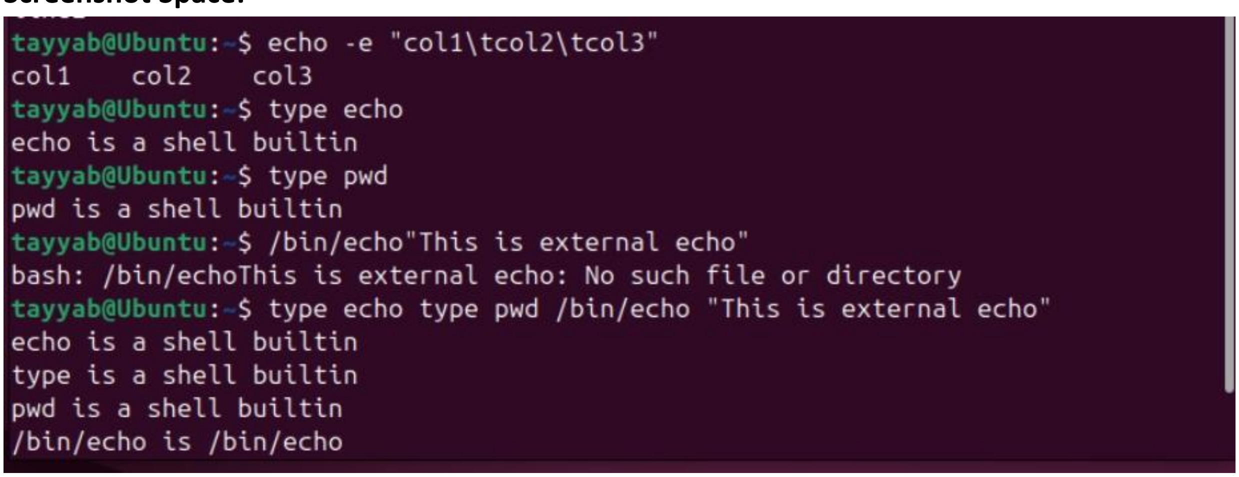
Commands:

type echo

type pwd

/bin/echo "This is external echo"

Screenshot Space:



TOPIC 5: Finding Command Locations (which)

Description:

The which command searches the system PATH and shows the exact file location of external commands.

Commands:

which mv

which cp

Screenshot Space:

TOPIC 6: Alias Management (create, view, remove)

Description:

Aliases allow you to create shortcuts or default options for commands.

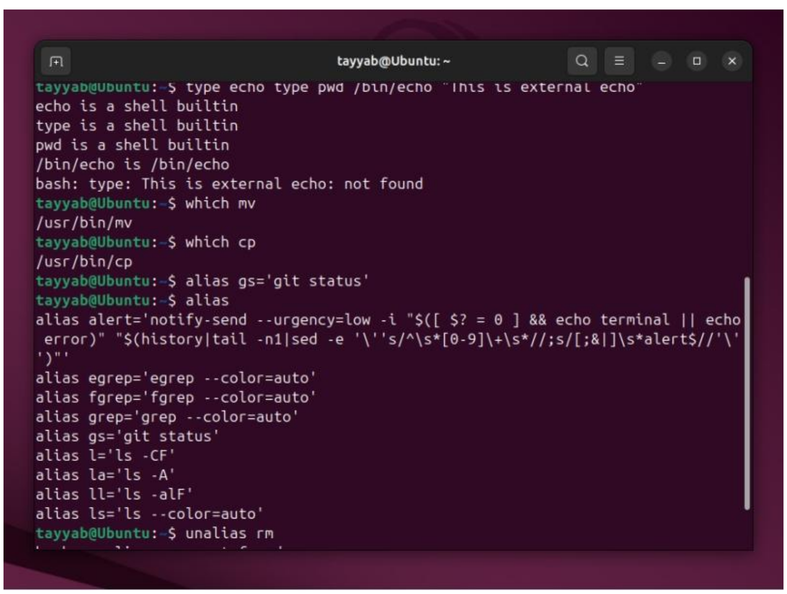
Commands:

alias gs='git status'

alias

unalias rm

Screenshot Space:



All Main Topic Names (Only the 6 Topics)

1. Command Line Scan, Arguments & White Space Handling
2. Quoting Mechanisms (Single & Double Quotes)
3. echo with Escape Sequences (echo -e)
4. Built-in vs External Commands (type & full path)
5. Finding Command Locations (which)
6. Alias Management (create, view, remove)

Lab Number #08:

UNIX PROCESS MANAGEMENT

Objective:

To understand and practice process management in UNIX/Linux systems using Ubuntu terminal commands.

Introduction

A process is a program in a running state. Each process is uniquely identified by a Process ID (PID. Processes can run in the foreground or background, and users can manage them using various commands.

Commands to Use in Ubuntu Linux Terminal

`ps` - Display Running Processes

Description:Shows the currently running processes.

Example:

```bash
```

ps

``

`ps -f` - Detailed Process List

Description: Displays detailed information about running processes, including PPID (Parent Process ID).

Example:

```bash

ps -f

``

## Run a Process in the Background

Description: Appending `&` to a command runs it in the background.

Example:

```bash

ls -l &

```

`jobs` - List Background and Suspended Jobs
```

Description: Shows all background and suspended processes.

Example:

```bash
jobs
```

`fg` - Bring a Background Process to the Foreground

Description: Brings a background job to the foreground using its job number.

Example:

```bash

fg %1

`bg` - Run a Suspended Process in the Background

Description:Resumes a suspended job in the background.

Example:

```bash
bg %1
```

7. `kill` - Terminate a Process
```

Description: Stops a process using its PID or job number.

Example:

```bash
kill %1
```

or

```bash
kill 1234
```

`sleep` - Create a Timed Process

Description:Pauses execution for a specified number of seconds. Useful for testing foreground/background behavior.

Example:

```bash
sleep 30
```

Lab Tasks

Task 1: Foreground Process

1. Open the terminal.

2. Run the following command:

```bash
sleep 60
```

3. Observe that the terminal is unresponsive until the process completes.

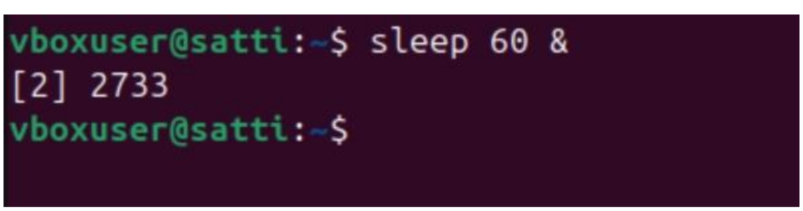
Task 2: Background Process

1. Run the same command in the background:

```bash
sleep 60 &
```

2. Note the job number and PID displayed.

3. Use `jobs` to see the background process.



```
vboxuser@satti:~$ jobs
[1]-  Done                    sleep 60
[2]+  Done                    sleep 60
vboxuser@satti:~$
```

Task 3: Process Listing

1. Use `ps` to list all running processes.

2. Use `ps -f` to see detailed information.

```
vboxuser@satti:~$ ps
    PID TTY          TIME CMD
   2720 pts/0    00:00:00 bash
   2747 pts/0    00:00:00 ps
vboxuser@satti:~$ ps -f
UID          PID    PPID  C STIME TTY          TIME CMD
vboxuser    2720    2713  0 10:33 pts/0    00:00:00 bash
vboxuser    2748    2720 99 10:38 pts/0    00:00:00 ps -f
```

Task 4: Process Management

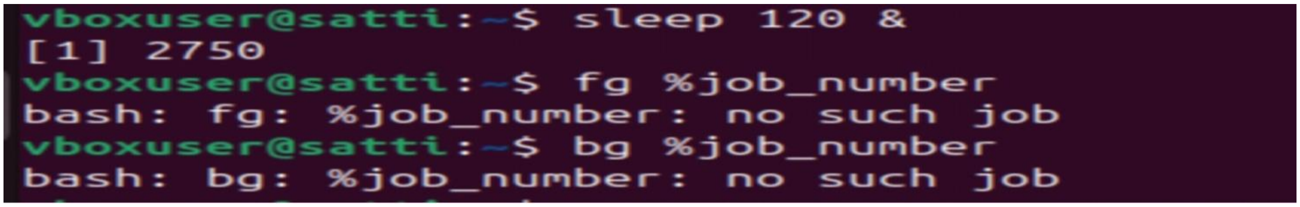
1. Start a process in the background:

```bash
sleep 120 &
```

2. Bring it to the foreground using `fg %job_number`.

3. Suspend it with `Ctrl + Z`.

4. Resume it in the background with `bg %job_number`.



Conclusion

In this lab, we learned how to:

- Differentiate between foreground and background processes.

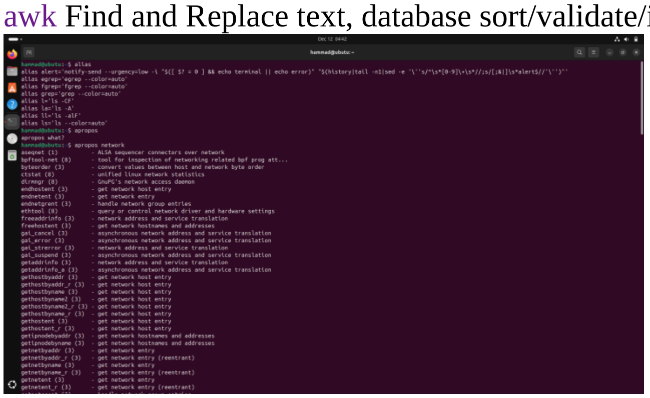
- Use commands like `ps`, `jobs`, `fg`, `bg`, and `kill` to manage processes.

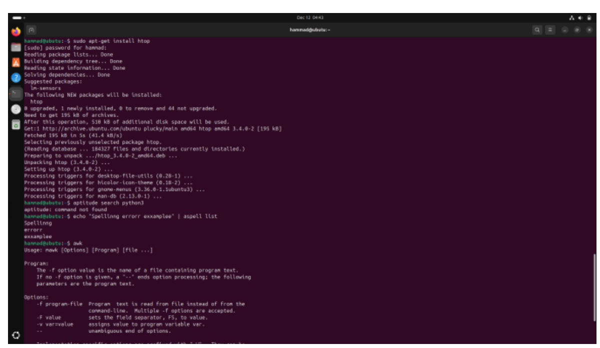
- Understand the relationship between parent and child processes.

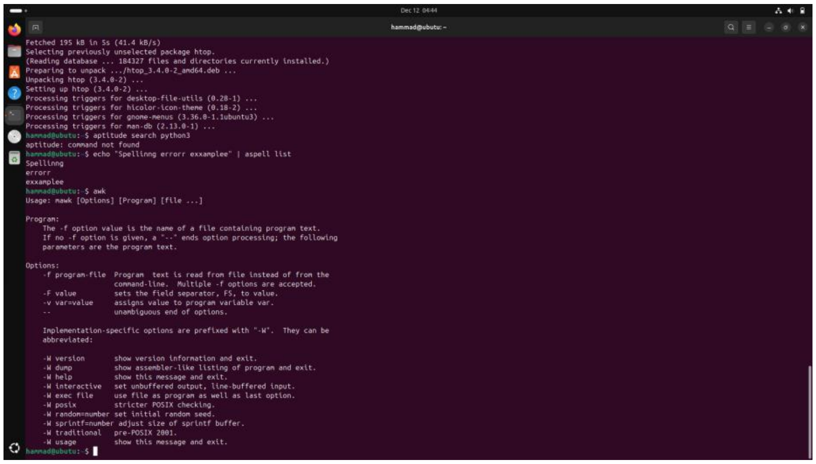
These skills are essential for effective system administration and process control in UNIX-like operating systems.

A-Z Index of the Bash command line for Linux

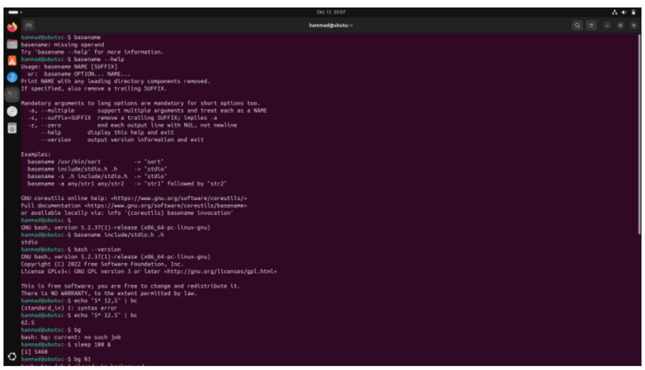
alias Create an alias
apropos Search Help manual pages (man -k)
apt-get Search for and install software packages (Debian/Ubuntu)
aptitude Search for and install software packages (Debian/Ubuntu)
aspell Spell Checker
awk Find and Replace text, database sort/validate/index

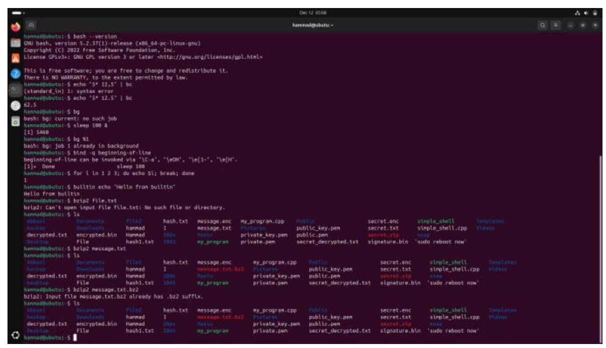






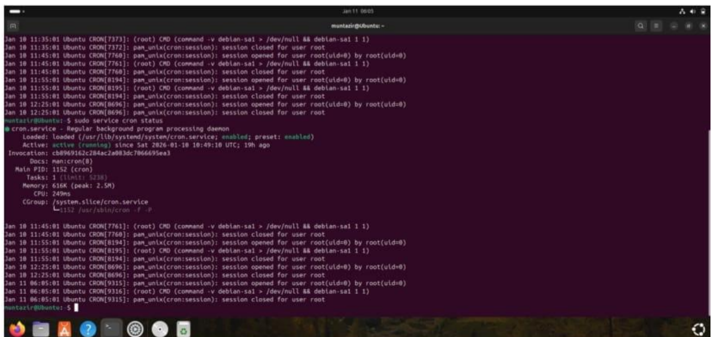
basename Strip directory and suffix fromfilenames
bash GNU Bourne-Again SHell
bc Arbitrary precision calculatorlanguage
bg Send to background
bind Set or display readline key and function bindings
break Exit from a loop
builtin Run a shell builtin
bzip2 Compress or decompress named file(s)

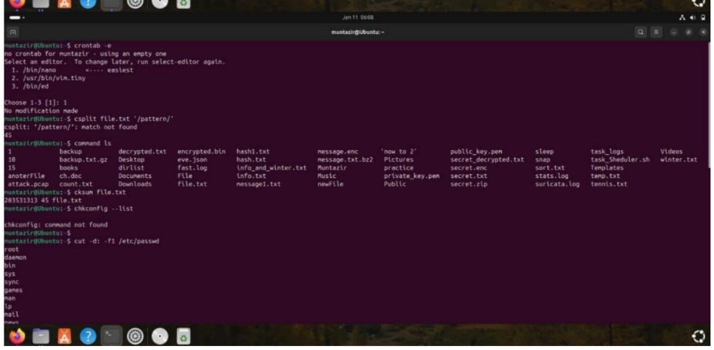


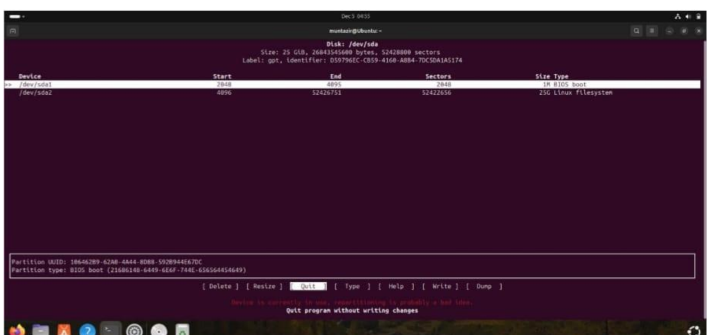


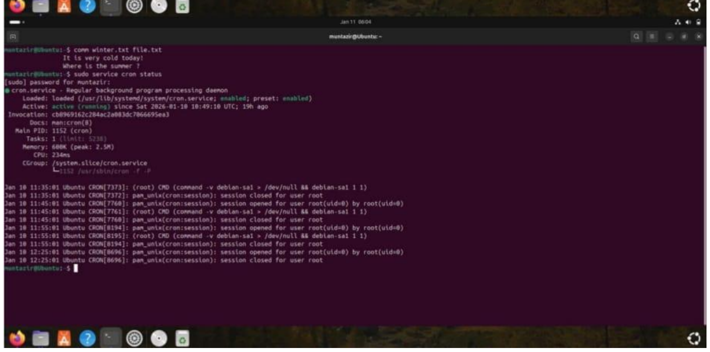
cal Display a calendar
case Conditionally perform a command
cat Concatenate and print (display) the content offiles
cd Change Directory
cfdisk Partition table manipulator for Linux
chattr Change file attributes on a Linux filesystem
chgrp Change group ownership
chmod Change access permissions
chown Change file owner andgroup
chroot Run a command with a different rootdirectory
chkconfig System services(runlevel)
cksum Print CRC checksum and bytecounts
clear Clear terminal screen
cmp Compare two files

comm Compare two sorted files line by line
command Run a command - ignoring shell functions
continue Resume the next iteration of a loop
cp Copy one or more files to another location
cron Daemon to execute scheduled commands
crontab Schedule a command to run at a later time
csplit Split a file into context-determined pieces
curl Transfer data from or to a server
cut Divide a file into several parts

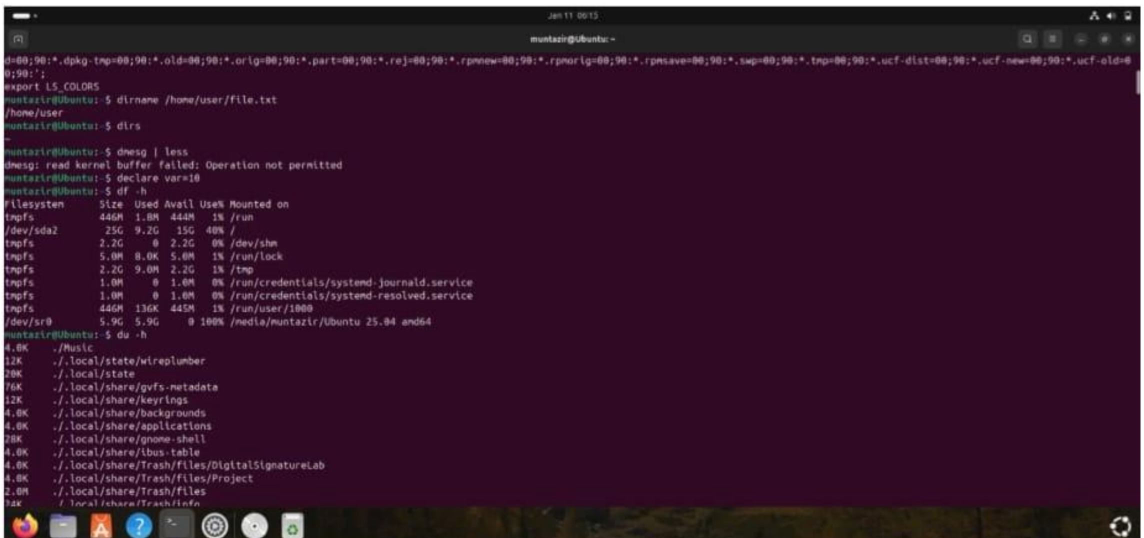


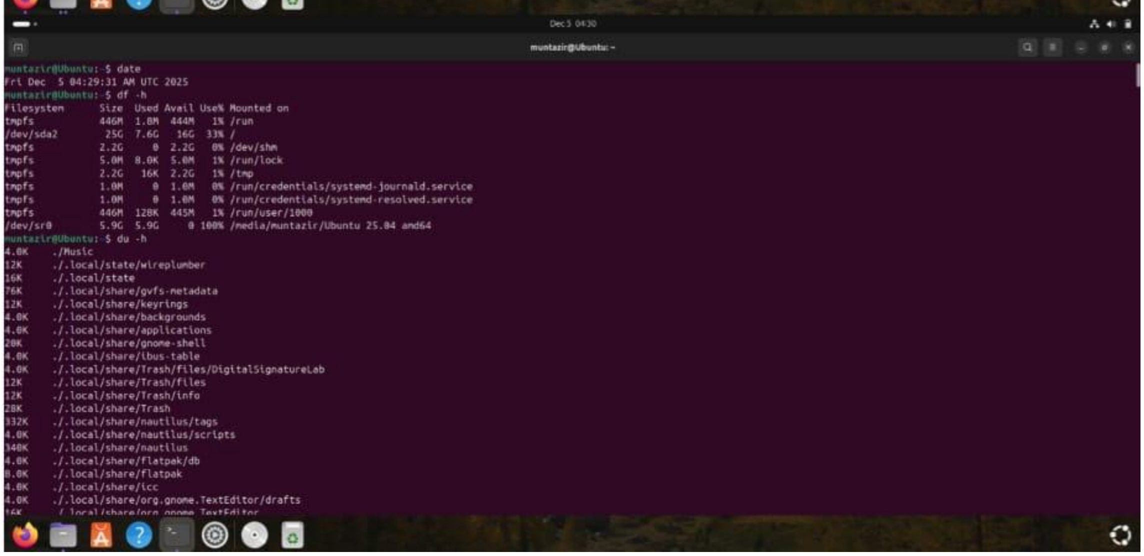


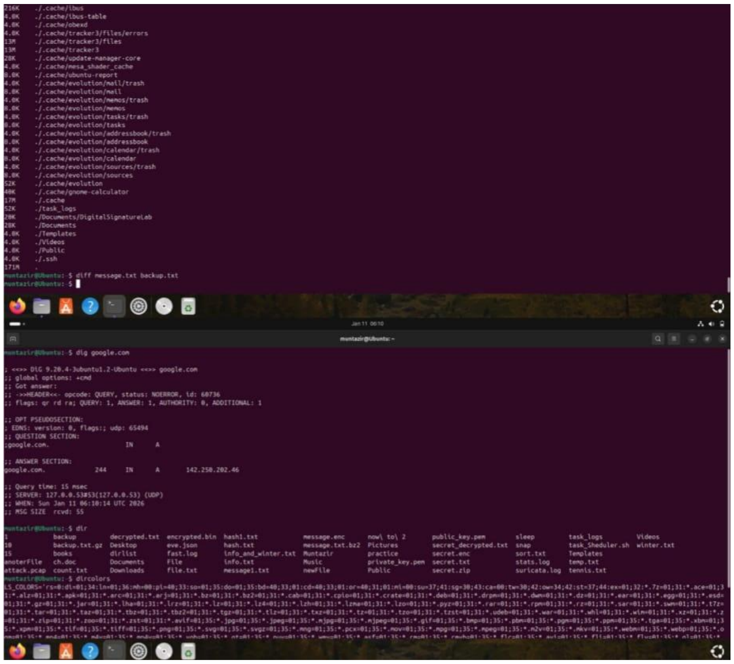




date Display or change the date &time
dc Desk Calculator
dd Convert and copy a file, write disk headers, bootrecords
ddrescue Data recovery tool
declare Declare variables and give them attributes•
df Display free disk space
diff Display the differences between twofiles
diff3 Show differences among three files
dig DNS lookup
dir Briefly list directorycontents
dircolors Colour setup for `ls'
dirname Convert a full pathname to just a path
dirs Display list of remembered directories
dmesg Print kernel & driver messages
du Estimate file space usage







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0. echo — Display message on screen.

echo "Hello World"

1. egrep — Search file(s) for lines that match an extended expression.

echo -e "cat\ndog\ncow" | egrep "dog|cow"

2. eject — Eject removable media.

eject /dev/cdrom

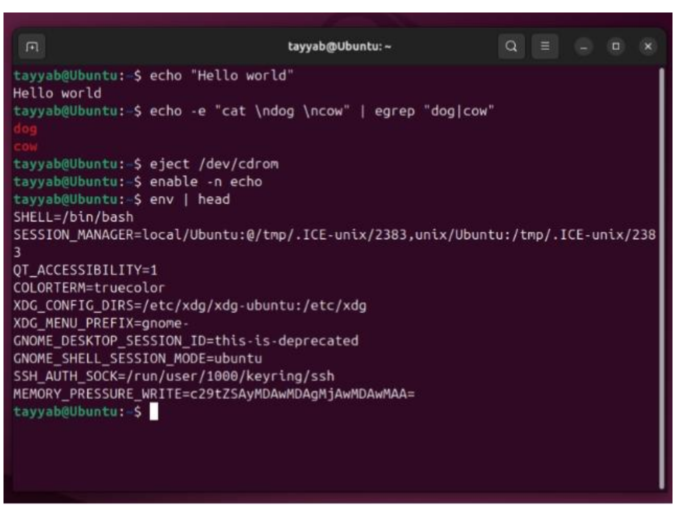
3. enable — Enable or disable builtin shell commands.

enable -n echo

4. env — Display environment variables.

Env

Screenshot :



5. eval — Evaluate several commands or arguments.
cmd="ls"; eval $cmd

6. exec — Execute a command by replacing the shell.
exec ls

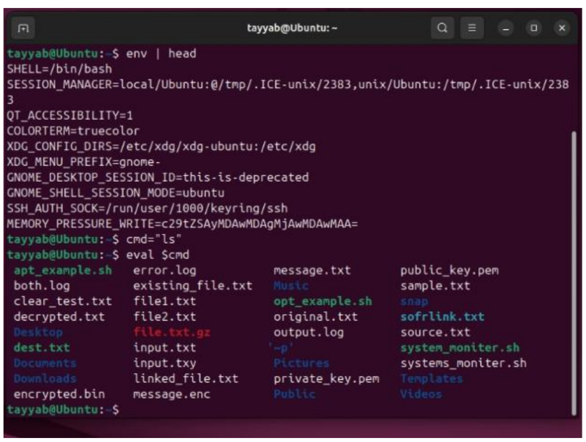
7. exit — Exit the shell.
exit

8. expect — Automate interactive terminal applications.
expect -c 'spawn ftp localhost; expect "Name"; send "user\r"'

9. expand — Convert tabs to spaces.
expand file.txt

10. export — Set an environment variable.
export MYVAR=10

11. expr — Evaluate expressions.
expr 10 + 5



12. false — Do nothing and return failure status.
false

13. fdformat — Low-level format a floppy disk.
sudo fdformat /dev/fd0

14. fdisk — Partition table manipulator for Linux.
sudo fdisk -l

15. fg — Send background job to foreground.
fg %1

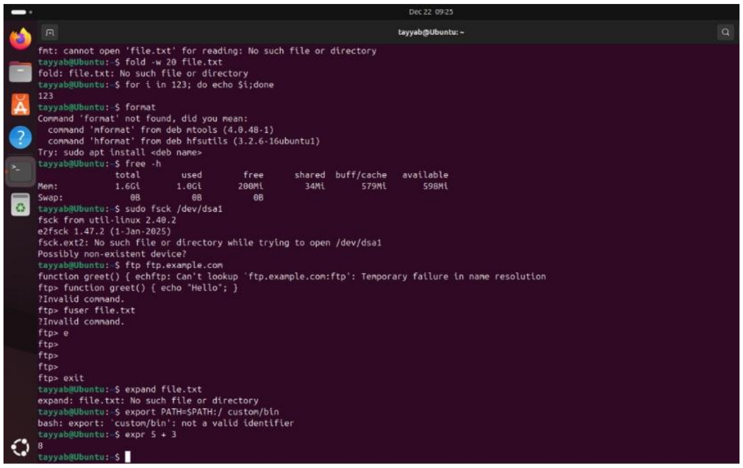
16. fgrep — Search file(s) for a fixed string.
echo "hello world" | fgrep world

17. file — Determine file type.
file test.txt

18. find — Search for files matching criteria.

find . -name "*.txt"

Screenshot :



19. fmt — Reformat paragraph text.

fmt file.txt

20. fold — Wrap text to a specified width.

fold -w 20 file.txt

21. for — Loop through values and execute commands.

for i in 1 2 3; do echo $i; done

22. format — Format disks or tapes.

format

23. free — Display memory usage.

free -h

24. fsck — Check and repair file system.

sudo fsck /dev/sda1

25. ftp — Transfer files using FTP protocol.

ftp ftp.example.com

26. function — Define a shell function.

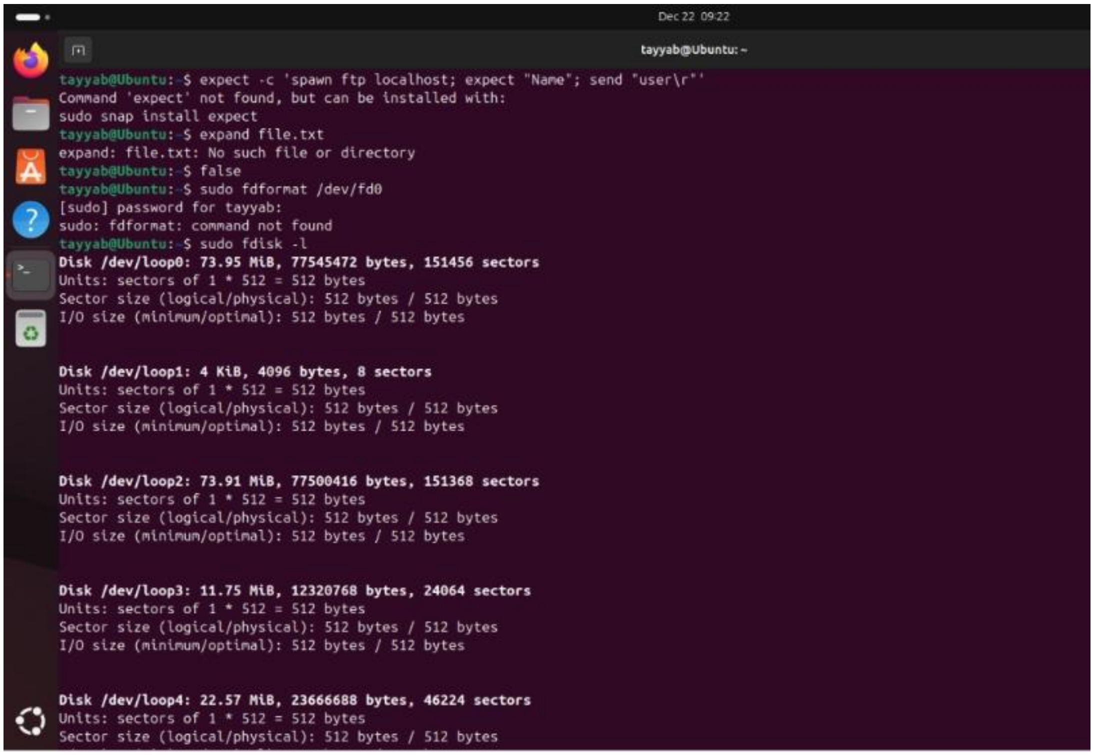
myfunc(){ echo "Hello"; }

myfunc

27. fuser — Identify process using a file.

fuser file.txt

Screenshot :



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1. **gawk** — Pattern scanning and text processing.

echo "Hello World" | gawk '{print $1}'

2. **getopts** — Parse command-line options in shell scripts.

while getopts ":a" opt; do case $opt in a) echo "Option -a";; esac; done < <(echo "-a")

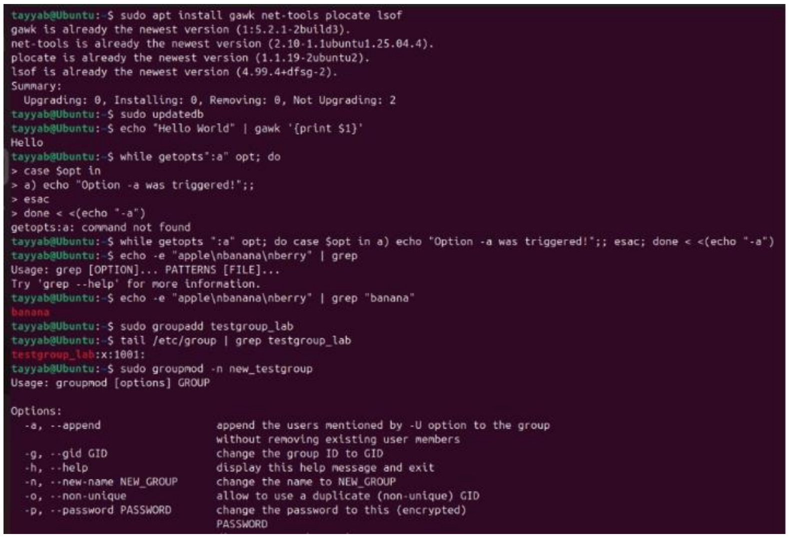
3. **grep** — Search text using patterns.

echo -e "apple\nbanana" | grep banana

4. **groupadd** — Create a new user group.

sudo groupadd testgroup_lab

Screenshot :

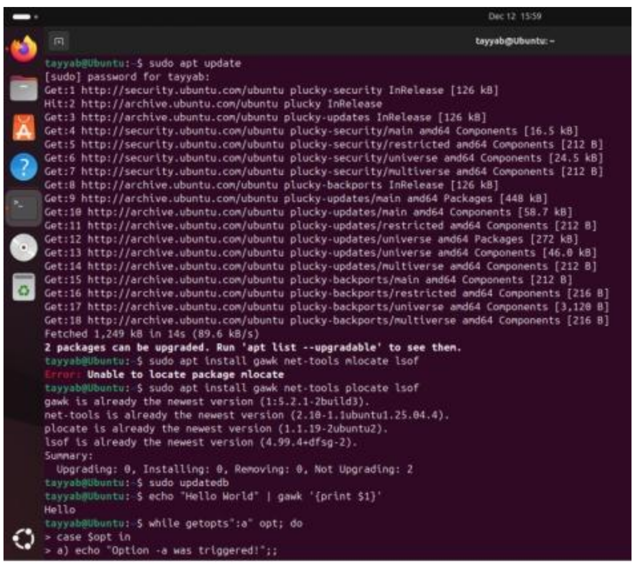


Installation / Update Space:

sudo apt update

sudo apt install gawk

Screenshot :



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5. **groupmod** — Modify an existing group.

sudo groupmod -n new_testgroup testgroup_lab

6. **groups** — Show groups of current user.

   groups

7. **groupdel** — Delete a group.

   sudo groupdel new_testgroup

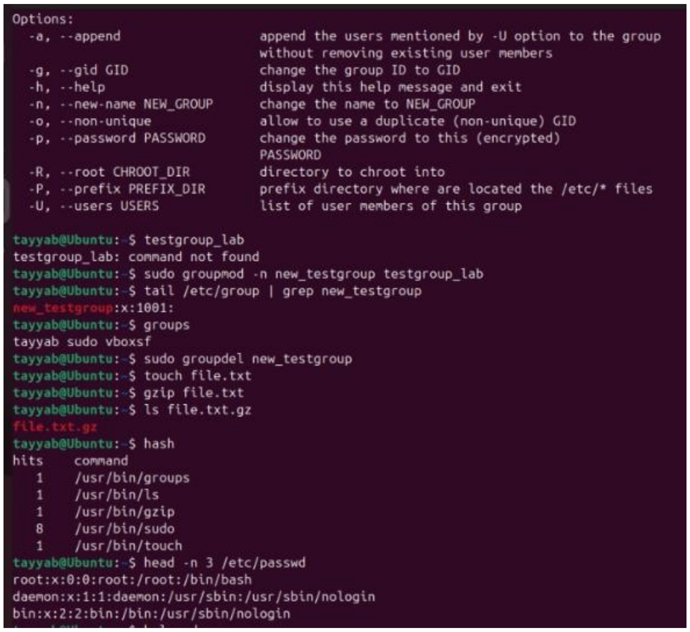
8. **gzip** — Compress files.

   touch file.txt && gzip file.txt

9. **hash** — Display command hash table.

hash

Screenshot):



```
Options:
  -a, --append                  append the users mentioned by -U option to the group
                                without removing existing user members
  -g, --gid GID                 change the group ID to GID
  -h, --help                    display this help message and exit
  -n, --new-name NEW_GROUP      change the name to NEW_GROUP
  -o, --non-unique              allow to use a duplicate (non-unique) GID
  -p, --password PASSWORD       change the password to this (encrypted)
                                PASSWORD
  -R, --root CHROOT_DIR         directory to chroot into
  -P, --prefix PREFIX_DIR       prefix directory where are located the /etc/* files
  -U, --users USERS             list of user members of this group

tayyab@Ubuntu:~$ testgroup_lab
testgroup_lab: command not found
tayyab@Ubuntu:~$ sudo groupmod -n new_testgroup testgroup_lab
tayyab@Ubuntu:~$ tail /etc/group | grep new_testgroup
new_testgroup:x:1001:
tayyab@Ubuntu:~$ groups
tayyab sudo vboxsf
tayyab@Ubuntu:~$ sudo groupdel new_testgroup
tayyab@Ubuntu:~$ touch file.txt
tayyab@Ubuntu:~$ gzip file.txt
tayyab@Ubuntu:~$ ls file.txt.gz
file.txt.gz
tayyab@Ubuntu:~$ hash
hits    command
   1    /usr/bin/groups
   1    /usr/bin/ls
   1    /usr/bin/gzip
   8    /usr/bin/sudo
   1    /usr/bin/touch
tayyab@Ubuntu:~$ head -n 3 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
```

---

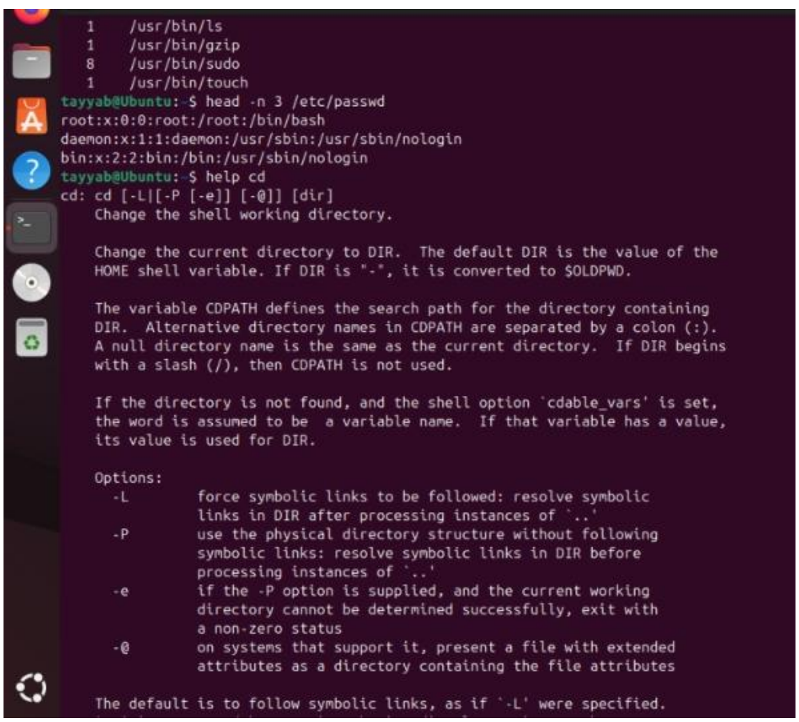
10. **head** — Show first lines of a file.

head -n 3 /etc/passwd

11. **help** — Get help for shell built-ins.

help cd

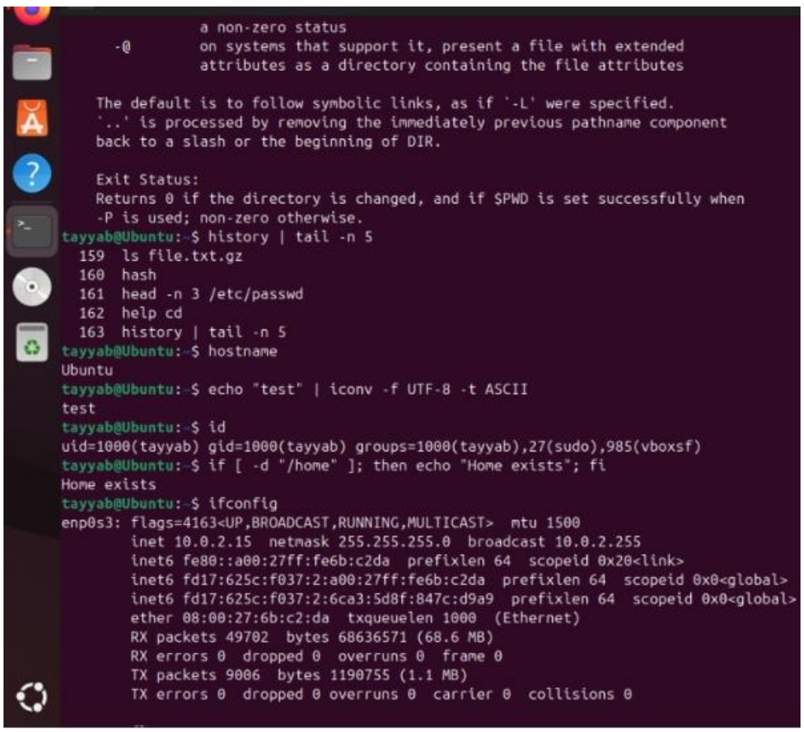
Screenshot :



```
   1    /usr/bin/ls
   1    /usr/bin/gzip
   8    /usr/bin/sudo
   1    /usr/bin/touch
tayyab@Ubuntu:~$ head -n 3 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
tayyab@Ubuntu:~$ help cd
cd: cd [-L|[-P [-e]] [-@]] [dir]
    Change the shell working directory.

    Change the current directory to DIR.  The default DIR is the value of the
    HOME shell variable. If DIR is "-", it is converted to $OLDPWD.

    The variable CDPATH defines the search path for the directory containing
    DIR.  Alternative directory names in CDPATH are separated by a colon (:).
    A null directory name is the same as the current directory.  If DIR begins
    with a slash (/), then CDPATH is not used.

    If the directory is not found, and the shell option `cdable_vars' is set,
    the word is assumed to be  a variable name.  If that variable has a value,
    its value is used for DIR.

    Options:
      -L        force symbolic links to be followed: resolve symbolic
                links in DIR after processing instances of `..'
      -P        use the physical directory structure without following
                symbolic links: resolve symbolic links in DIR before
                processing instances of `..'
      -e        if the -P option is supplied, and the current working
                directory cannot be determined successfully, exit with
                a non-zero status
      -@        on systems that support it, present a file with extended
                attributes as a directory containing the file attributes

    The default is to follow symbolic links, as if `-L' were specified.
```

12. **history** — View command history.
history | tail -n 5
13. **hostname** — Show system hostname.
hostname
14. **iconv** — Character encoding conversion.
echo "test" | iconv -f UTF-8 -t ASCII
15. **id** — Display user and group IDs.
id
16. **if** — Conditional execution in shell scripts.
if [-d /home]; then echo "Home exists"; fi
17. **ifconfig** — Display network interface configuration.
ifconfig

Screenshot :



Installation / Update Space:

sudo apt install net-tools

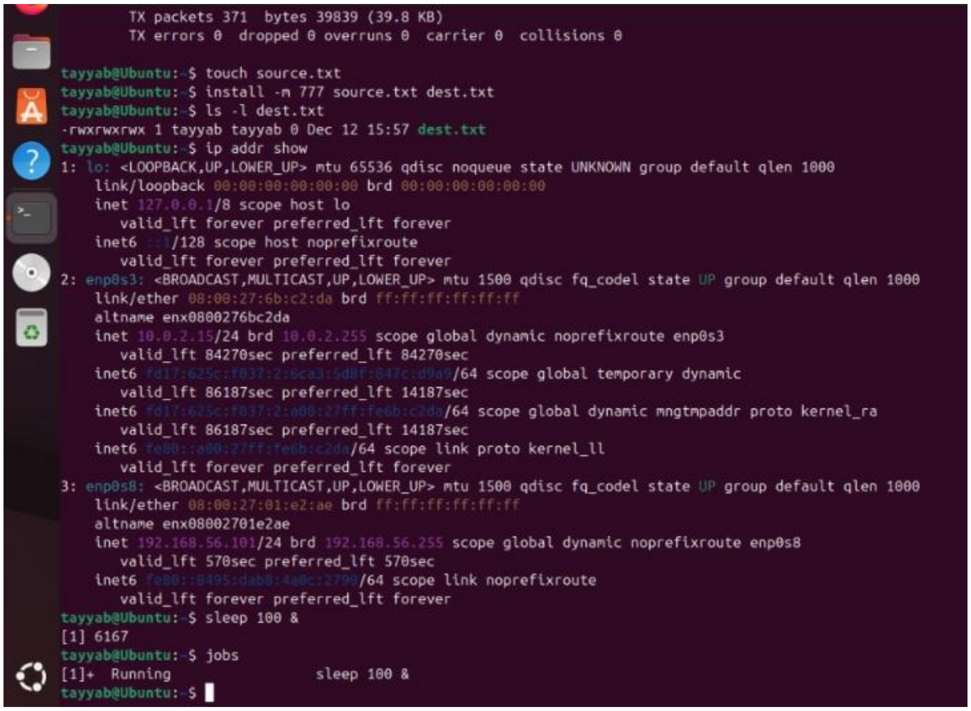
---

18. **install** — Copy files and set ownership and permissions.
touch source.txt
install -m 755 source.txt dest.txt
ls -l dest.txt
19. **ip** — Show and manage network interfaces.
ip addr show
20. **jobs** — Display background jobs.

sleep 60 &

jobs

Screenshot :



---

21. **join** — Join lines of two files on a common field.

echo "1 A" > f1.txt

echo "1 B" > f2.txt

join f1.txt f2.txt

22. **kill** — Terminate a process using PID.

sleep 100 &

kill $!

23. **killall** — Terminate processes by name.

sleep 100 &

killall sleep

24. **let** — Perform arithmetic operations.

let sum=5+5

echo $sum

25. **link** — Create a hard link to a file.

touch original.txt

link original.txt hardlink.txt

ls -l hardlink.txt

26. **ln** — Create a symbolic link.

ln -s original.txt softlink.txt

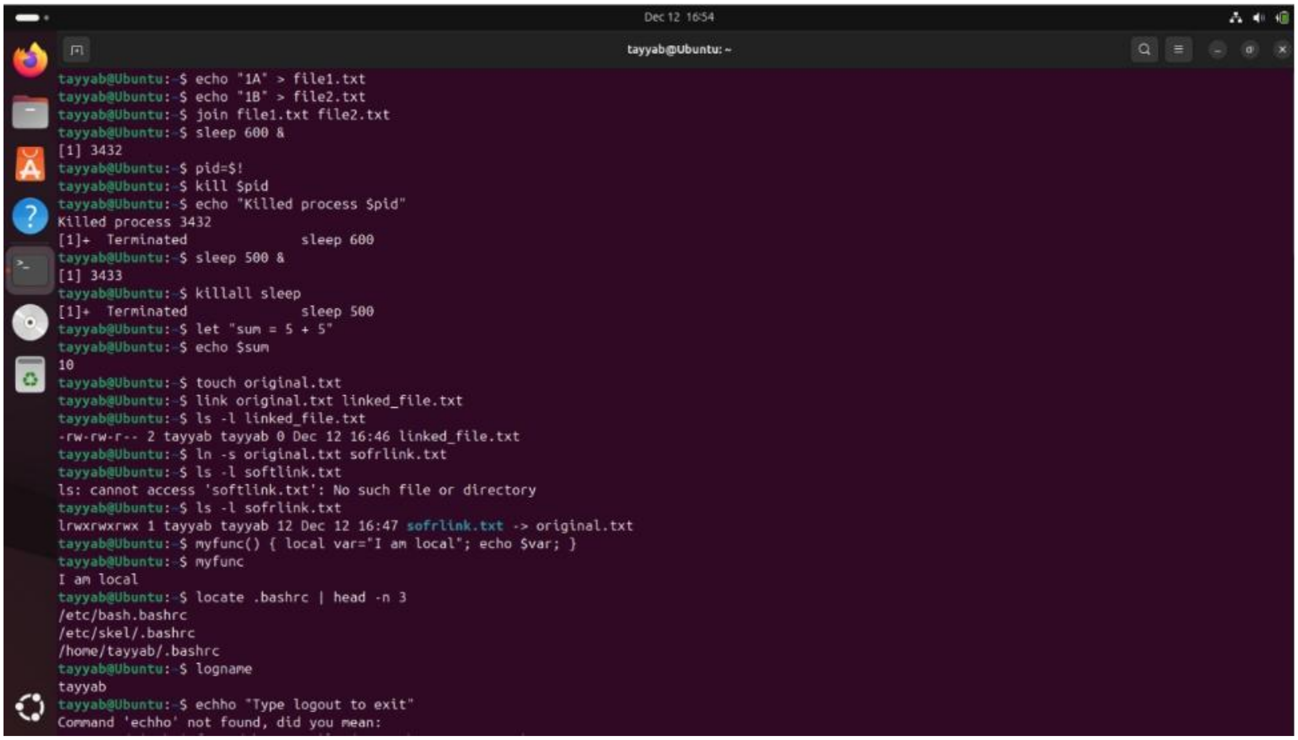
ls -l softlink.txt

27. **local** — Declare local variables in functions.

myfunc(){ local x=10; echo $x; }

myfunc

Screenshot :



---

28. **locate** — Search for files using database.

locate .bashrc | head

29. **logname** — Display login name of the user.

logname

30. **logout** — Exit login shell.

echo "Type logout to exit shell"

31. **look** — Display dictionary words starting with pattern.

look bas /usr/share/dict/words | head

32. **lpc** — Printer control program.

/usr/sbin/lpc status

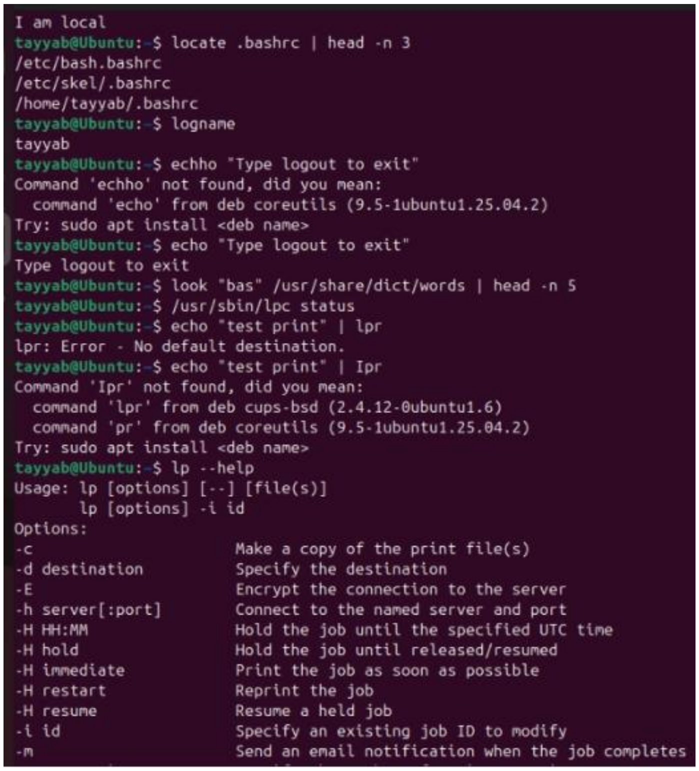
33. **lpr** — Send file to printer.

echo "Test print" | lpr

34. **lprint** — Print files (lp alternative).

lp --help

Screenshot :



```
I am local
tayyab@Ubuntu:~$ locate .bashrc | head -n 3
/etc/bash.bashrc
/etc/skel/.bashrc
/home/tayyab/.bashrc
tayyab@Ubuntu:~$ logname
tayyab
tayyab@Ubuntu:~$ echho "Type logout to exit"
Command 'echho' not found, did you mean:
  command 'echo' from deb coreutils (9.5-1ubuntu1.25.04.2)
Try: sudo apt install <deb name>
tayyab@Ubuntu:~$ echo "Type logout to exit"
Type logout to exit
tayyab@Ubuntu:~$ look "bas" /usr/share/dict/words | head -n 5
tayyab@Ubuntu:~$ /usr/sbin/lpc status
tayyab@Ubuntu:~$ echo "test print" | lpr
lpr: Error - No default destination.
tayyab@Ubuntu:~$ echo "test print" | Ipr
Command 'Ipr' not found, did you mean:
  command 'lpr' from deb cups-bsd (2.4.12-0ubuntu1.6)
  command 'pr' from deb coreutils (9.5-1ubuntu1.25.04.2)
Try: sudo apt install <deb name>
tayyab@Ubuntu:~$ lp --help
Usage: lp [options] [--] [file(s)]
       lp [options] -i id
Options:
-c                      Make a copy of the print file(s)
-d destination          Specify the destination
-E                      Encrypt the connection to the server
-h server[:port]        Connect to the named server and port
-H HH:MM                Hold the job until the specified UTC time
-H hold                 Hold the job until released/resumed
-H immediate            Print the job as soon as possible
-H restart              Reprint the job
-H resume               Resume a held job
-i id                   Specify an existing job ID to modify
-m                      Send an email notification when the job completes
```

---

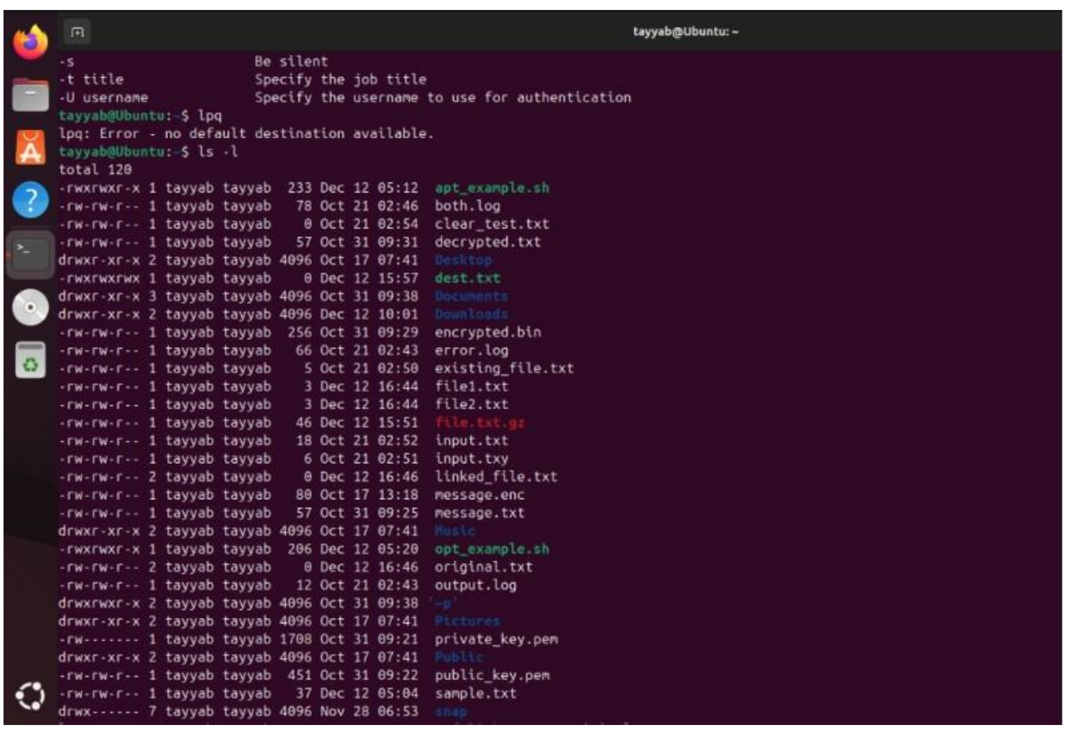
35. **lpq** — Display printer queue status.

lpq

36. **ls** — List directory contents.

ls -l

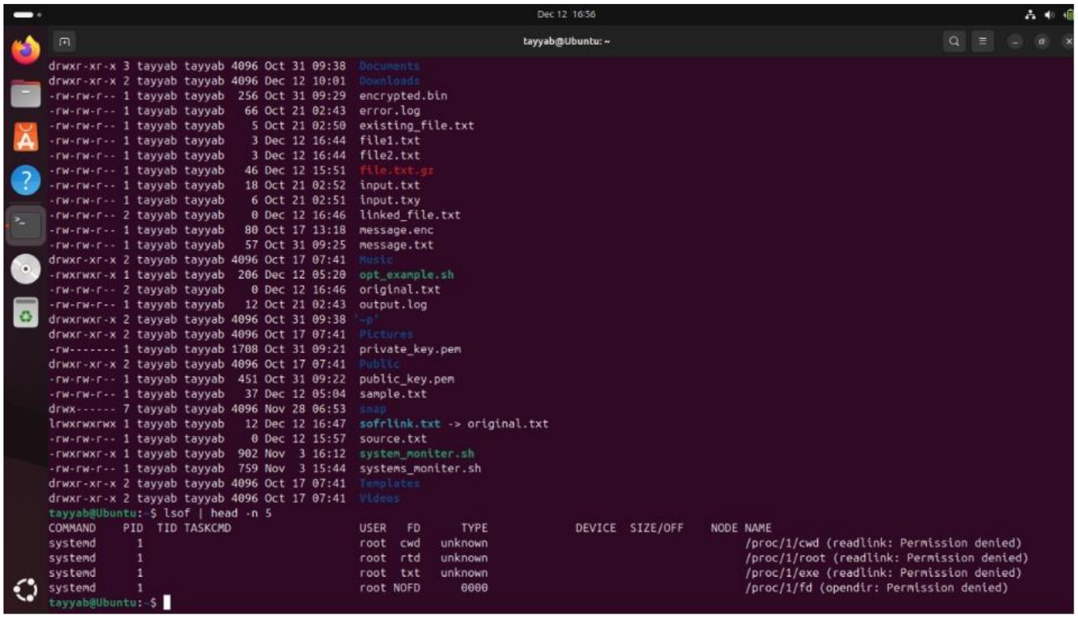
Screenshot :



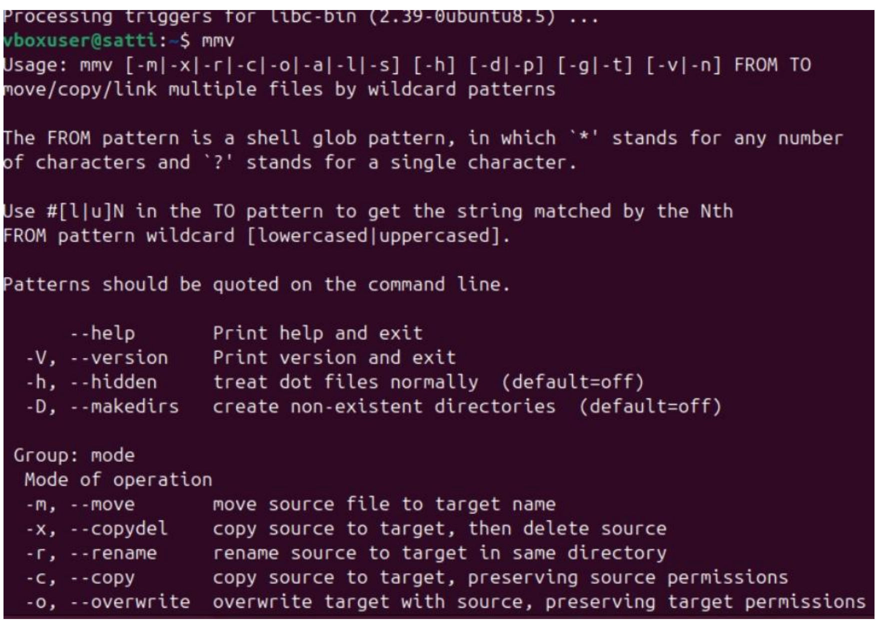
```
tayyab@Ubuntu: ~
-s                      Be silent
-t title                Specify the job title
-U username             Specify the username to use for authentication
tayyab@Ubuntu:~$ lpq
lpq: Error - no default destination available.
tayyab@Ubuntu:~$ ls -l
total 120
-rwxrwxr-x 1 tayyab tayyab  233 Dec 12 05:12  apt_example.sh
-rw-rw-r-- 1 tayyab tayyab   78 Oct 21 02:46  both.log
-rw-rw-r-- 1 tayyab tayyab    0 Oct 21 02:54  clear_test.txt
-rw-rw-r-- 1 tayyab tayyab   57 Oct 31 09:31  decrypted.txt
drwxr-xr-x 2 tayyab tayyab 4096 Oct 17 07:41  Desktop
-rwxrwxrwx 1 tayyab tayyab    0 Dec 12 15:57  dest.txt
drwxr-xr-x 3 tayyab tayyab 4096 Oct 31 09:38  Documents
drwxr-xr-x 2 tayyab tayyab 4096 Dec 12 10:01  Downloads
-rw-rw-r-- 1 tayyab tayyab  256 Oct 31 09:29  encrypted.bin
-rw-rw-r-- 1 tayyab tayyab   66 Oct 21 02:43  error.log
-rw-rw-r-- 1 tayyab tayyab    5 Oct 21 02:50  existing_file.txt
-rw-rw-r-- 1 tayyab tayyab    3 Dec 12 16:44  file1.txt
-rw-rw-r-- 1 tayyab tayyab    3 Dec 12 16:44  file2.txt
-rw-rw-r-- 1 tayyab tayyab   46 Dec 12 15:51  file.txt.gz
-rw-rw-r-- 1 tayyab tayyab   18 Oct 21 02:52  input.txt
-rw-rw-r-- 1 tayyab tayyab    6 Oct 21 02:51  input.txy
-rw-rw-r-- 2 tayyab tayyab    0 Dec 12 16:46  linked_file.txt
-rw-rw-r-- 1 tayyab tayyab   80 Oct 17 13:18  message.enc
-rw-rw-r-- 1 tayyab tayyab   57 Oct 31 09:25  message.txt
drwxr-xr-x 2 tayyab tayyab 4096 Oct 17 07:41  Music
-rwxrwxr-x 1 tayyab tayyab  206 Dec 12 05:20  opt_example.sh
-rw-rw-r-- 2 tayyab tayyab    0 Dec 12 16:46  original.txt
-rw-rw-r-- 1 tayyab tayyab   12 Oct 21 02:43  output.log
drwxrwxr-x 2 tayyab tayyab 4096 Oct 31 09:38  '-p'
drwxr-xr-x 2 tayyab tayyab 4096 Oct 17 07:41  Pictures
-rw------- 1 tayyab tayyab 1708 Oct 31 09:21  private_key.pem
drwxr-xr-x 2 tayyab tayyab 4096 Oct 17 07:41  Public
-rw-rw-r-- 1 tayyab tayyab  451 Oct 31 09:22  public_key.pem
-rw-rw-r-- 1 tayyab tayyab   37 Dec 12 05:04  sample.txt
drwx------ 7 tayyab tayyab 4096 Nov 28 06:53  snap
```

37. **lsof** — List open files and associated processes.

lsof | head -n 5



MMV:



```
Group: mode
 Mode of operation
 -m, --move       move source file to target name
 -x, --copydel    copy source to target, then delete source
 -r, --rename     rename source to target in same directory
 -c, --copy       copy source to target, preserving source permissions
 -o, --overwrite  overwrite target with source, preserving target permissions
 -l, --hardlink   link target name to source file
 -s, --symlink    symlink target name to source file

Group: delete
 How to handle file deletions and overwrites
 -d, --force      perform file deletes and overwrites without confirmation
 -p, --protect    treat file deletes and overwrites as errors

Group: erroneous
 How to handle erroneous actions
 -g, --go         skip any erroneous actions
 -t, --terminate  erroneous actions are treated as errors

Group: report
 Reporting actions
 -v, --verbose    report all actions performed
 -n, --dryrun     only report which actions would be performed
boxuser@satti:~$
```

Mtools:

```
vboxuser@satti:~$ mtools
Supported commands:
mattrib, mbadblocks, mcat, mcd, mcopy, mdel, mdeltree, mdir
mdoctorfat, mdu, mformat, minfo, mlabel, mmd, mmount, mpartition
mrd, mread, mmove, mren, mshowfat, mshortname, mtoolstest, mtype
mwrite, mzip
```

MMM

```
                              My traceroute  [v0.95]
atti (::1) -> localhost (::1)                          2025-12-20T13:27:59+0000
Keys:  Help   Display mode   Restart statistics   Order of fields   quit
                                       Packets               Pings
Host                                 Loss%   Snt   Last   Avg  Best  Wrst StDev
1. localhost6.localdomain6            0.0%    13    0.1   0.1   0.0   0.2   0.0
```

MV:

```
vboxuser@satti:~$ mv --help
Usage: mv [OPTION]... [-T] SOURCE DEST
  or:  mv [OPTION]... SOURCE... DIRECTORY
  or:  mv [OPTION]... -t DIRECTORY SOURCE...
Rename SOURCE to DEST, or move SOURCE(s) to DIRECTORY.

Mandatory arguments to long options are mandatory for short options too.
      --backup[=CONTROL]       make a backup of each existing destination file
  -b                           like --backup but does not accept an argument
      --debug                  explain how a file is copied.  Implies -v
  -f, --force                  do not prompt before overwriting
  -i, --interactive            prompt before overwrite
  -n, --no-clobber             do not overwrite an existing file
If you specify more than one of -i, -f, -n, only the final one takes effect.
      --no-copy                do not copy if renaming fails
      --strip-trailing-slashes  remove any trailing slashes from each SOURCE
                                 argument
  -S, --suffix=SUFFIX          override the usual backup suffix
  -t, --target-directory=DIRECTORY  move all SOURCE arguments into DIRECTORY
  -T, --no-target-directory    treat DEST as a normal file
  --update[=UPDATE]            control which existing files are updated;
                                 UPDATE={all,none,older(default)}.  See below
  -u                           equivalent to --update[=older]
  -v, --verbose                explain what is being done
  -Z, --context                set SELinux security context of destination
```

N COMMANDS:

Nc:

```
vboxuser@satti:~$ nc
usage: nc [-46CDdFhklNnrStUuvZz] [-I length] [-i interval] [-M ttl]
          [-m minttl] [-O length] [-P proxy_username] [-p source_port]
          [-q seconds] [-s sourceaddr] [-T keyword] [-V rtable] [-W recvlimit]
          [-w timeout] [-X proxy_protocol] [-x proxy_address[:port]]
          [destination] [port]
```

\NETSTAT:

```
vboxuser@satti:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 satti:60406             ubuntu-archive-mir:http TIME_WAIT
udp        0      0 satti:bootpc            _gateway:bootps         ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  2      [ ]         DGRAM                    16579
unix  3      [ ]         STREAM     CONNECTED     16523
unix  3      [ ]         STREAM     CONNECTED     8578
unix  3      [ ]         STREAM     CONNECTED     8514
unix  3      [ ]         STREAM     CONNECTED     15597
unix  3      [ ]         STREAM     CONNECTED     15421
unix  3      [ ]         STREAM     CONNECTED     16760    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     16716
unix  3      [ ]         STREAM     CONNECTED     16194
unix  3      [ ]         STREAM     CONNECTED     16605    /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     8711
unix  3      [ ]         STREAM     CONNECTED     15412    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     8782
unix  2      [ ]         DGRAM      CONNECTED     8732
unix  3      [ ]         STREAM     CONNECTED     16525    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     15600
```

```
vboxuser@satti: ~
unix  3      [ ]         STREAM     CONNECTED     8782
unix  2      [ ]         DGRAM      CONNECTED     8732
unix  3      [ ]         STREAM     CONNECTED     16525    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     15600
unix  3      [ ]         STREAM     CONNECTED     8515     /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     16604
unix  3      [ ]         STREAM     CONNECTED     16589    /run/snapd-snap.socke
t
unix  3      [ ]         STREAM     CONNECTED     17151    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     9997
unix  3      [ ]         STREAM     CONNECTED     36028
unix  3      [ ]         STREAM     CONNECTED     16526    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     15405
unix  3      [ ]         STREAM     CONNECTED     13378    /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     16608
unix  3      [ ]         STREAM     CONNECTED     8712     /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     16759
unix  3      [ ]         STREAM     CONNECTED     16588
unix  3      [ ]         STREAM     CONNECTED     16179    /run/user/1000/gvfsd/
socket-MSOMfI3e
unix  3      [ ]         STREAM     CONNECTED     13908    /run/dbus/system_bus_
```

vboxuser@satti: ~

```
unix  3      [ ]         STREAM     CONNECTED     13908    /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     16909
unix  3      [ ]         STREAM     CONNECTED     16297    /run/user/1000/waylan
d-0
unix  2      [ ]         DGRAM      CONNECTED     8781
unix  3      [ ]         STREAM     CONNECTED     15408
unix  3      [ ]         STREAM     CONNECTED     13907
unix  3      [ ]         STREAM     CONNECTED     16786    /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     16761
unix  3      [ ]         STREAM     CONNECTED     16592
unix  3      [ ]         STREAM     CONNECTED     16195    /run/systemd/journal/
stdout
unix  2      [ ]         DGRAM                    8518
unix  3      [ ]         STREAM     CONNECTED     16528
unix  3      [ ]         STREAM     CONNECTED     16914    /run/user/1000/at-spi
/bus
unix  3      [ ]         STREAM     CONNECTED     8668     /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     17152
unix  3      [ ]         STREAM     CONNECTED     16557
unix  2      [ ]         DGRAM                    15385
unix  3      [ ]         STREAM     CONNECTED     13726    /run/systemd/journal/
stdout
```

```
unix  3      [ ]         STREAM     CONNECTED     16557
unix  2      [ ]         DGRAM      CONNECTED     15385
unix  3      [ ]         STREAM     CONNECTED     13726    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     12449    /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     16609    /run/user/1000/pulse/
native
unix  3      [ ]         STREAM     CONNECTED     16522
unix  3      [ ]         STREAM     CONNECTED     15601    /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     15387    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     8949
unix  3      [ ]         STREAM     CONNECTED     8519
unix  3      [ ]         DGRAM      CONNECTED     13569
unix  3      [ ]         STREAM     CONNECTED     17149
unix  3      [ ]         STREAM     CONNECTED     8757     /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     16558    /run/user/1000/at-spi
/bus
unix  3      [ ]         STREAM     CONNECTED     13504
unix  3      [ ]         STREAM     CONNECTED     8579
unix  3      [ ]         STREAM     CONNECTED     36029
unix  3      [ ]         STREAM     CONNECTED     16530    /run/user/1000/bus
```

Dec 20 13:44

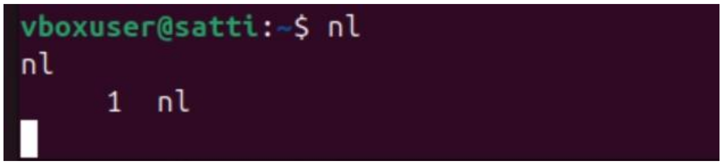
vboxuser@satti: ~

```
unix  3      [ ]         STREAM     CONNECTED     16530    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     15386
unix  3      [ ]         STREAM     CONNECTED     9998     /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     16881    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     12352
unix  3      [ ]         STREAM     CONNECTED     17153    /run/user/1000/waylan
d-0
unix  3      [ ]         STREAM     CONNECTED     16854    /run/user/1000/pipewi
re-0
unix  3      [ ]         STREAM     CONNECTED     12447
unix  3      [ ]         STREAM     CONNECTED     15573    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     15406    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     16907    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     8950     /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     8520     /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     17154
unix  3      [ ]         STREAM     CONNECTED     13505    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     15768    /home/vboxuser/.cache
/ibus/dbus-bkrvxQpK
unix  3      [ ]         STREAM     CONNECTED     16355
```

NICE:

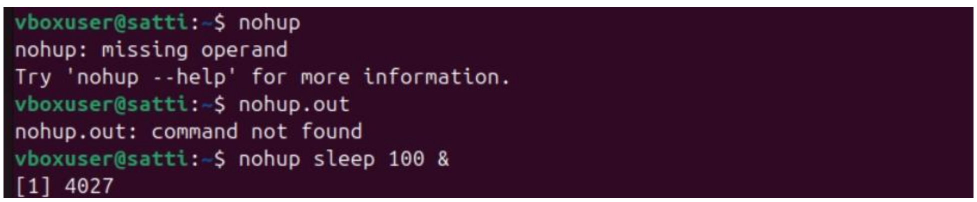
```
vboxuser@satti:~$ nice
0
```

NL:



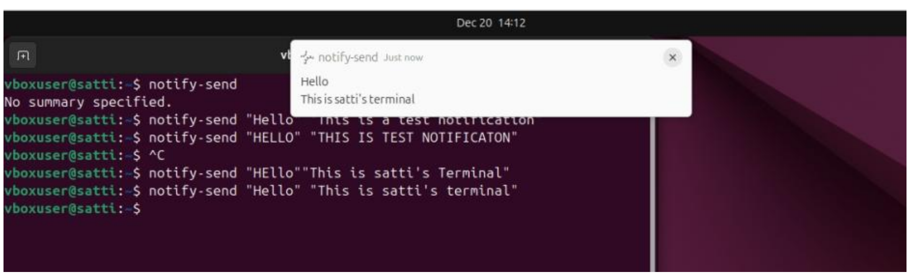
```
vboxuser@satti:~$ nl
nl
     1	nl
```

NOHUP:



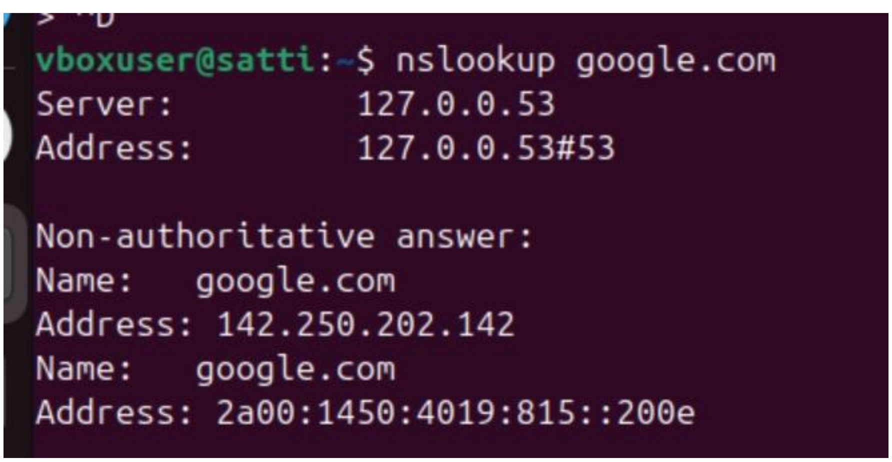
```
vboxuser@satti:~$ nohup
nohup: missing operand
Try 'nohup --help' for more information.
vboxuser@satti:~$ nohup.out
nohup.out: command not found
vboxuser@satti:~$ nohup sleep 100 &
[1] 4027
```

NOTIFY:




```
vboxuser@satti:~$ notify-send
No summary specified.
vboxuser@satti:~$ notify-send "Hello" "This is a test notification"
vboxuser@satti:~$ notify-send "HELLO" "THIS IS TEST NOTIFICATON"
vboxuser@satti:~$ ^C
vboxuser@satti:~$ notify-send "HEllo""This is satti's Terminal"
vboxuser@satti:~$ notify-send "Hello" "This is satti's terminal"
vboxuser@satti:~$
```

NS-Look UP:



```
vboxuser@satti:~$ nslookup google.com
Server:		127.0.0.53
Address:	127.0.0.53#53

Non-authoritative answer:
Name:	google.com
Address: 142.250.202.142
Name:	google.com
Address: 2a00:1450:4019:815::200e
```

O commands:

OP:

```
vboxuser@satti:~$ op
op: command not found
vboxuser@satti:~$ alias op='xdg-open'
vboxuser@satti:~$
vboxuser@satti:~$
```

OPEN:

```
vboxuser@satti:~$ open
xdg-open - opens a file or URL in the user's preferred application

Synopsis

xdg-open { file | URL }

xdg-open { --help | --manual | --version }

Use 'man xdg-open' or 'xdg-open --manual' for additional info.
vboxuser@satti:~$ open youtube.com
gio: file:///home/vboxuser/youtube.com: Error when getting information for file
"/home/vboxuser/youtube.com": No such file or directory
vboxuser@satti:~$ open google.com
gio: file:///home/vboxuser/google.com: Error when getting information for file "
/home/vboxuser/google.com": No such file or directory
```

P COMMANDS:

PASSWD:

```
vboxuser@satti:~$ passwd
Changing password for vboxuser.
Current password:
New password:
BAD PASSWORD: The password is shorter than 8 characters
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/sys
tematic
New password:
Retype new password:
passwd: password updated successfully
```

PASTE:

```
vboxuser@satti:~$ paste
vboxuser@satti:~$ type paste
paste is hashed (/usr/bin/paste)
vboxuser@satti:~$ alias paste
bash: alias: paste: not found
vboxuser@satti:~$ unalias paste
bash: unalias: paste: not found
vboxuser@satti:~$
```

```
vboxuser@satti:~$ pgrep bash
2686
vboxuser@satti:~$ pgrep firefox
vboxuser@satti:~$ pgrep python
vboxuser@satti:~$
```

```
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=66 ttl=255
time=51.0 ms

d
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=67 ttl=255
time=50.0 ms
dd
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=68 ttl=255
time=50.4 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=69 ttl=255
time=49.9 ms
^X64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=70 ttl=25
5 time=53.2 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=71 ttl=255
time=50.4 ms
^C
--- youtube.com ping statistics ---
71 packets transmitted, 70 received, 1.40845% packet loss, time 70603ms
rtt min/avg/max/mdev = 45.331/410.036/4120.404/770.318 ms, pipe 5
```

PING:

```
vboxuser@satti:~$ ping youtube.com
PING youtube.com (142.250.201.238) 56(84) bytes of data.
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=1 ttl=255 t
ime=1218 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=2 ttl=255 t
ime=1206 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=3 ttl=255 t
ime=209 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=5 ttl=255 t
ime=1302 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=6 ttl=255 t
ime=749 ms
64 bytes from pnfjra-ap-in-f14.1e100.net (142.250.201.238): icmp_seq=7 ttl=255 t
```

PKILL:

```
vboxuser@satti:~$ pkill 5MB testfile
pkill: only one pattern can be provided
Try `pkill --help' for more information.
vboxuser@satti:~$ pkill
pkill: no matching criteria specified
Try `pkill --help' for more information.
vboxuser@satti:~$ pkill chrome
vboxuser@satti:~$ pkill firefox
vboxuser@satti:~$
```

POPD:

```
vboxuser@satti:~$ popd
bash: popd: directory stack empty
vboxuser@satti:~$ popd /tmp
bash: popd: /tmp: invalid argument
popd: usage: popd [-n] [+N | -N]
vboxuser@satti:~$ popd +1
bash: popd: directory stack empty
vboxuser@satti:~$
```

PR:

```
vboxuser@satti:~$ pr 5MB testfile
pr: 5MB: No such file or directory


2025-12-19 13:39                    testfile                    Page 1
```

PRINTCAP:

```
vboxuser@satti:~$ printcap
Command 'printcap' not found, did you mean:
  command 'printcal' from deb argyll (3.1.0+repack-1)
Try: sudo apt install <deb name>
vboxuser@satti:~$ ls -la /etc/printcap
lrwxrwxrwx 1 root root 18 Dec 21 13:42 /etc/printcap -> /run/cups/printcap
```

PRINTENV:

```
vboxuser@satti:~$ printenv
SHELL=/bin/bash
SESSION_MANAGER=local/satti:@/tmp/.ICE-unix/1890,unix/satti:/tmp/.ICE-unix/1890
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
MEMORY_PRESSURE_WRITE=c29tZSAyMDAwMDAgMjAwMDAwMAA=
XMODIFIERS=@im=ibus
DESKTOP_SESSION=ubuntu
GTK_MODULES=gail:atk-bridge
PWD=/home/vboxuser
LOGNAME=vboxuser
XDG_SESSION_DESKTOP=ubuntu
XDG_SESSION_TYPE=wayland
SYSTEMD_EXEC_PID=1934
XAUTHORITY=/run/user/1000/.mutter-Xwaylandauth.9K1UH3
HOME=/home/vboxuser
USERNAME=vboxuser
```

```
USERNAME=vboxuser
IM_CONFIG_PHASE=1
LANG=en_US.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd
=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:e
x=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:
*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=0
1;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01
;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz
2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*
.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;
31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif
=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;
35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif
=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35
:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.web
p=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:
*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01
;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=
01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.f
lac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;
36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xsp
```

```
SHLVL=1
GSM_SKIP_SSH_AGENT_WORKAROUND=true
QT_IM_MODULE=ibus
XDG_RUNTIME_DIR=/run/user/1000
DEBUGINFOD_URLS=https://debuginfod.ubuntu.com
XDG_DATA_DIRS=/usr/share/ubuntu:/usr/share/gnome:/usr/local/share/:/usr/share/:/
var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/us
r/local/games:/snap/bin:/snap/bin
GDMSESSION=ubuntu
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
_=/usr/bin/printenv
```

```
f=00;36:*~=00;90:*#=00;90:*.bak=00;90:*.crdownload=00;90:*.dpkg-dist=00;90:*.dpk
g-new=00;90:*.dpkg-old=00;90:*.dpkg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00
;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:*.swp=00;90:*.tmp
=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
XDG_CURRENT_DESKTOP=ubuntu:GNOME
MEMORY_PRESSURE_WATCH=/sys/fs/cgroup/user.slice/user-1000.slice/user@1000.servic
e/session.slice/org.gnome.Shell@wayland.service/memory.pressure
VTE_VERSION=7600
WAYLAND_DISPLAY=wayland-0
GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/077e92ab_9bf0_4641_85cf_fdb42b5
a3661
GNOME_SETUP_DISPLAY=:1
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=vboxuser
GNOME_TERMINAL_SERVICE=:1.97
CLUTTER_DISABLE_MIPMAPPED_TEXT=1
DISPLAY=:0
SHLVL=1
```

PRINTF:

```
vboxuser@satti:~$ printf
printf: usage: printf [-v var] format [arguments]
```

PS:

```
vboxuser@satti:~$ ps
    PID TTY          TIME CMD
   2686 pts/0    00:00:00 bash
   4185 pts/0    00:00:00 ps
```

PUSHD:

```
vboxuser@satti:~$ pushd
bash: pushd: no other directory
vboxuser@satti:~$ pushd 5MB testfile
bash: pushd: too many arguments
```

PV:

```
vboxuser@satti:~$ pv
^D0.00  B 0:00:30 [0.00  B/s] [<=>
1.00  B 0:00:33 [30.3miB/s] [ <=>
```

```
vboxuser@satti:~$ pv
0.00  B 0:00:11 [0.00  B/s] [<=>
```

PWD:

```
vboxuser@satti:~$ pwd
/home/vboxuser
```

Q COMMANDS:

QUOTA:

```
vboxuser@satti:~$ quota
vboxuser@satti:~$ quota
vboxuser@satti:~$
```

```
-R, --exclude-root         exclude root when checking all filesystems
-F, --format=formatname    check quota files of specific format
-a, --all                  check all filesystems
-h, --help                 display this message and exit
-V, --version              display version information and exit

Bugs to jack@suse.cz
vboxuser@satti:~$
```

```
vboxuser@satti:~$ quota
vboxuser@satti:~$ quota
vboxuser@satti:~$ quotacheck
Bad number of arguments.
Utility for checking and repairing quota files.
quotacheck [-gucbfinvdmMR] [-F <quota-format>] filesystem|-a

-u, --user                check user files
-g, --group               check group files
-c, --create-files        create new quota files
-b, --backup              create backups of old quota files
-f, --force               force check even if quotas are enabled
-i, --interactive         interactive mode
-n, --use-first-dquot     use the first copy of duplicated structure
-v, --verbose             print more information
-d, --debug               print even more messages
-m, --no-remount          do not remount filesystem read-only
-M, --try-remount         try remounting filesystem read-only,
                          continue even if it fails
-R, --exclude-root        exclude root when checking all filesystems
-F, --format=formatname   check quota files of specific format
-a, --all                 check all filesystems
-h, --help                display this message and exit
-V, --version             display version information and exit
```

R COMMANDS:

RAM:

```
vboxuser@satti:~$ ram
Command 'ram' not found, did you mean:
  command 'rem' from snap rem (0.17.0)
  command 'rar' from deb rar (2:6.23-1)
  command 'ra6' from deb ipv6toolkit (2.0+ds.1-2)
  command 'ra' from deb argus-client (1:3.0.8.2-6.2ubuntu1)
  command 'jam' from deb jam (2.6.1-2build2)
  command 'vam' from deb vim-addon-manager (0.5.10)
  command 'nam' from deb nam (1.15-6)
  command 'rmm' from deb mailutils-mh (1:3.16-1build1)
  command 'rmm' from deb mmh (0.4-6)
  command 'rmm' from deb nmh (1.8-1)
  command 'bam' from deb bam (0.5.1-3)
  command 'cam' from deb libcamera-tools (0.1.0-3ubuntu1)
  command 'rpm' from deb rpm (4.18.0+dfsg-1build2)
  command 'rm' from deb coreutils (9.4-3ubuntu6.1)
  command 'rem' from deb remind (04.02.08-1)
See 'snap info <snapname>' for additional versions.
vboxuser@satti:~$ ram free -h
Command 'ram' not found, did you mean:
  command 'rem' from snap rem (0.17.0)
  command 'rem' from deb remind (04.02.08-1)
```

```
vboxuser@satti:~$ rar

RAR 7.00   Copyright (c) 1993-2024 Alexander Roshal   26 Feb 2024
Trial version             Type 'rar -?' for help

Usage:     rar <command> -<switch 1> -<switch N> <archive> <files...>
               <@listfiles...> <path_to_extract/>

<Commands>
  a             Add files to archive
  c             Add archive comment
  ch            Change archive parameters
  cw            Write archive comment to file
  d             Delete files from archive
  e             Extract files without archived paths
  f             Freshen files in archive
  i[par]=<str>  Find string in archives
  k             Lock archive
  l[t[a],b]     List archive contents [technical[all], bare]
  m[f]          Move to archive [files only]
  p             Print file to stdout
  r             Repair archive
  rc            Reconstruct missing volumes
```

```
ds             Disable name sort for solid archive
dw             Wipe files after archiving
e[+]<attr>     Set file exclude and include attributes
ed             Do not add empty directories
ep             Exclude paths from names
ep1            Exclude base directory from names
ep3            Expand paths to full including the drive letter
ep4<path>      Exclude the path prefix from names
f              Freshen files
hp[password]   Encrypt both file data and headers
ht[b|c]        Select hash type [BLAKE2,CRC32] for file checksum
id[c,d,n,p,q]  Display or disable messages
ierr           Send all messages to stderr
ilog[name]     Log errors to file
inul           Disable all messages
isnd[-]        Control notification sounds
iver           Display the version number
k              Lock archive
kb             Keep broken extracted files
log[f][=name]  Write names to log file
m<0..5>        Set compression level (0-store...3-default...5-maximal)
mc<par>        Set advanced compression parameters
md[x]<n>[kmg]  Dictionary size in KB, MB or GB
me[par]        Set encryption parameters
```

```
mc<par>        Set advanced compression parameters
md[x]<n>[kmg]  Dictionary size in KB, MB or GB
me[par]        Set encryption parameters
ms[ext;ext]    Specify file types to store
mt<threads>    Set the number of threads
n<file>        Additionally filter included files
n@             Read additional filter masks from stdin
n@<list>       Read additional filter masks from list file
o[+|-]         Set the overwrite mode
oh             Save hard links as the link instead of the file
oi[0-4][:min]  Save identical files as references
ol[a,-]        Process symbolic links as the link [absolute paths, skip]
op<path>       Set the output path for extracted files
or             Rename files automatically
ow             Save or restore file owner and group
p[password]    Set password
qo[-|+]        Add quick open information [none|force]
r              Recurse subdirectories
r-             Disable recursion
r0             Recurse subdirectories for wildcard names only
rr[N]          Add data recovery record
rv[N]          Create recovery volumes
s[<N>,v[-],e]  Create solid archive
s-             Disable solid archiving
```

```
  sc<chr>[obj]  Specify the character set
  sfx[name]     Create SFX archive
  si[name]      Read data from standard input (stdin)
  sl<size>[u]   Process files with size less than specified
  sm<size>[u]   Process files with size more than specified
  t             Test files after archiving
  ta[mcao]<d>   Process files modified after <d> YYYYMMDDHHMMSS date
  tb[mcao]<d>   Process files modified before <d> YYYYMMDDHHMMSS date
  tk            Keep original archive time
  tl            Set archive time to latest file
  tn[mcao]<t>   Process files newer than <t> time
  to[mcao]<t>   Process files older than <t> time
  ts[m,c,a,p]   Save or restore time (modification, creation, access, preserve)
  u             Update files
  v<size>[u]    Create volumes with size in [bBkKmMgGtT] units
  ver[n]        File version control
  vp            Pause before each volume
  w<path>       Assign work directory
  x<file>       Exclude specified file
  x@            Read file names to exclude from stdin
  x@<list>      Exclude files listed in specified list file
  y             Assume Yes on all queries
  z[file]       Read archive comment from file
```

```
 rn            Rename archived files
 rr[N]         Add data recovery record
 rv[N]         Create recovery volumes
 s[name|-]     Convert archive to or from SFX
 t             Test archive files
 u             Update files in archive
 v[t[a],b]     Verbosely list archive contents [technical[all],bare]
 x             Extract files with full path

Switches>
 -             Stop switches scanning
 @[+]          Disable [enable] file lists
 ad[1,2]       Alternate destination path
 ag[format]    Generate archive name using the current date
 ai            Ignore file attributes
 am[s,r]       Archive name and time [save, restore]
 ap<path>      Set path inside archive
 as            Synchronize archive contents
 c-            Disable comments show
 cfg-          Disable read configuration
 cl            Convert names to lower case
 cu            Convert names to upper case
 df            Delete files after archiving
 dh            Open shared files
```

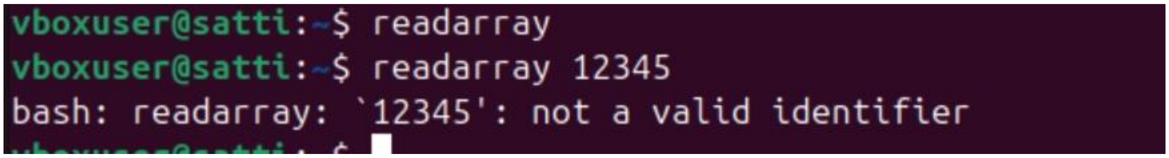
RCP:

```
vboxuser@satti:~$ rcp
Command 'rcp' not found, but can be installed with:
sudo apt install rsh-client
vboxuser@satti:~$ sudo apt install rsh-client
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package rsh-client is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

E: Package 'rsh-client' has no installation candidate
```

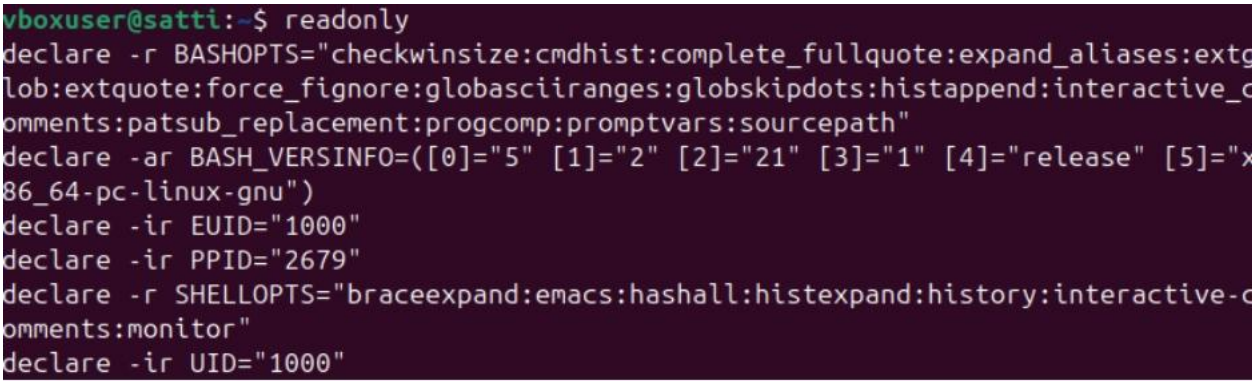
READ:

```
vboxuser@satti:~$ read
vboxuser@satti:~$ read 5MB testfile
bash: read: `5MB': not a valid identifier
vboxuser@satti:~$
```

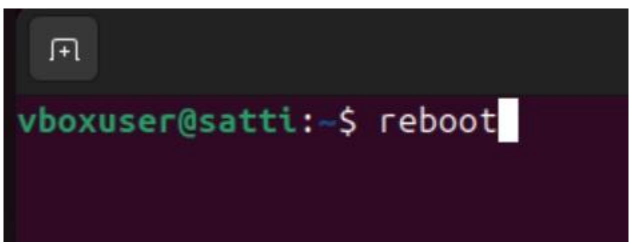
READ ARRAY:

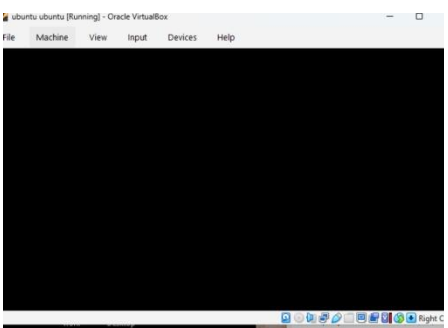


READONLY:

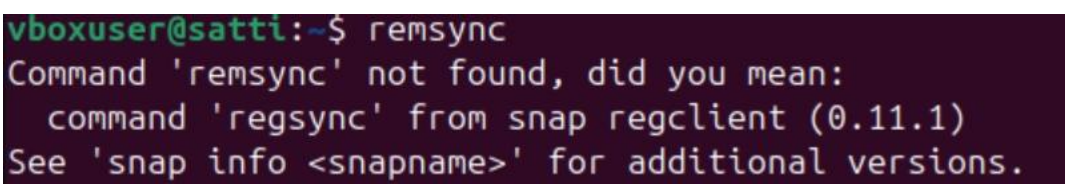


REBOOT:





REMSYNC:



RENAME:

```
vboxuser@satti:~$ rename
Usage:
    file-rename [ -h|-m|-V ] [ -v ] [ -0 ] [ -n ] [ -f ] [ -d ] [ -u [enc]]
    [ -e|-E perlexpr]*|perlexpr [ files ]
```

RENICE:

```
vboxuser@satti:~$ renice
renice: not enough arguments
Try 'renice --help' for more information.
vboxuser@satti:~$ renice -p
renice: not enough arguments
Try 'renice --help' for more information.
vboxuser@satti:~$ renice --help

Usage:
 renice [-n|--priority|--relative] <priority> [-p|--pid] <pid>...
 renice [-n|--priority|--relative] <priority>  -g|--pgrp <pgid>...
 renice [-n|--priority|--relative] <priority>  -u|--user <user>...

Alter the priority of running processes.

Options:
 -n <num>                specify the nice value
                           If POSIXLY_CORRECT flag is set in environment
                           then the priority is 'relative' to current
```

RETURN:

```
vboxuser@satti:~$ return
bash: return: can only `return' from a function or sourced script
vboxuser@satti:~$ return {n}
bash: return: {n}: numeric argument required
bash: return: can only `return' from a function or sourced script
vboxuser@satti:~$ return 300
bash: return: can only `return' from a function or sourced script
vboxuser@satti:~$
```

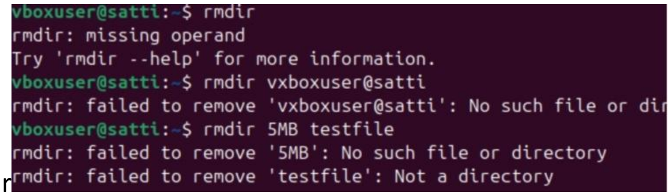
REV:

```
vboxuser@satti:~$ rev
vboxuser@satti:~$ rev 5MB testfile
rev: cannot open 5MB: No such file or directory
vboxuser@satti:~$
```

RM:

```
vboxuser@satti:~$ rm
rm: missing operand
Try 'rm --help' for more information.
vboxuser@satti:~$ rm -f
vboxuser@satti:~$ rm -i
rm: missing operand
Try 'rm --help' for more information.
vboxuser@satti:~$ rm "$file"
rm: cannot remove '': No such file or directory
vboxuser@satti:~$
```

RMDIR:



```
vboxuser@satti:~$ rmdir
rmdir: missing operand
Try 'rmdir --help' for more information.
vboxuser@satti:~$ rmdir vxboxuser@satti
rmdir: failed to remove 'vxboxuser@satti': No such file or dir
vboxuser@satti:~$ rmdir 5MB testfile
rmdir: failed to remove '5MB': No such file or directory
rmdir: failed to remove 'testfile': Not a directory
```

r

Dependencies/Installation: sudo apt install screen

Screenshot:

```
sohaib@sohaib:~$ sudo apt install screen
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
screen is already the newest version (4.9.1-1build1).
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
sohaib@sohaib:~$
```

Command Name: scp

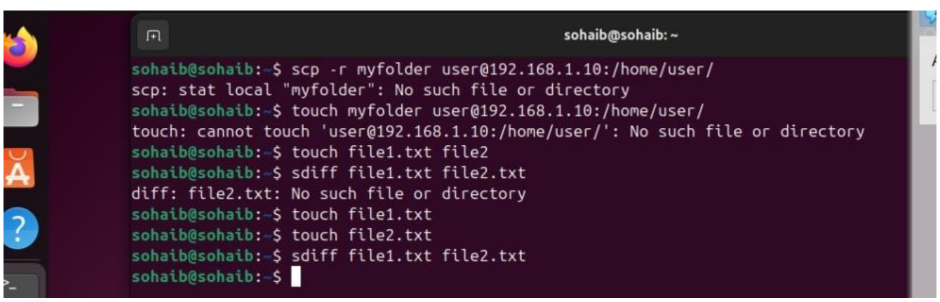
Command Syntax: scp [options] source_file user@host:destination_path

Description: Securely copies files between hosts over a network using SSH.

Dependencies/Installation: sudo apt install openssh-client

Screenshot:

Command Name: sdiff
Command Syntax: sdiff [options] file1 file2
Description: Compares two files side-by-side and shows differences.
Dependencies/Installation: Part of diffutils, typically pre-installed.
Screenshot:



Command Name: sed
Command Syntax: sed [options] 'script' input_file
Description: Parses and transforms text by applying scripts to an input stream.
Dependencies/Installation: Part of sed package, typically pre-installed.
Screenshot:

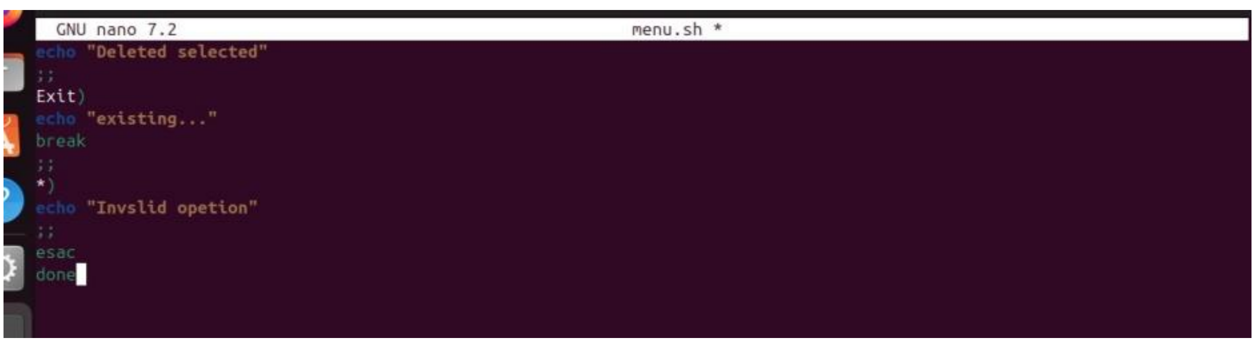
Command Name: select

Command Syntax: select name [in words ...]; do commands; done

Description: A shell built-in that generates simple text menus from a list.

Dependencies/Installation: Built into Bash, no installation needed.

Screenshot:



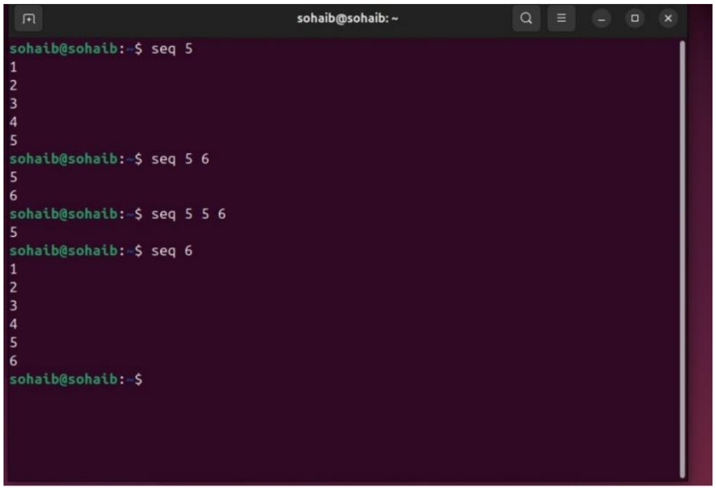
Command Name: seq

Command Syntax: seq [options] first last

Description: Generates a sequence of numbers.

Dependencies/Installation: Part of coreutils, typically pre-installed.

Screenshot:



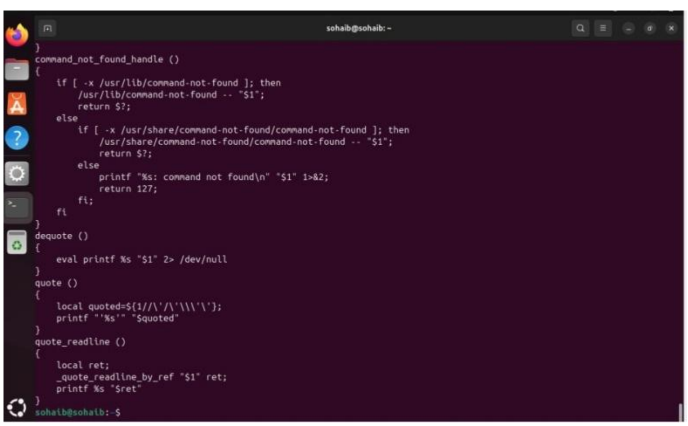
Command Name: set

Command Syntax: set [options] [arguments]

Description: Displays or sets shell environment variables and positional parameters.

Dependencies/Installation: Built into the shell, no installation needed.

Screenshot:



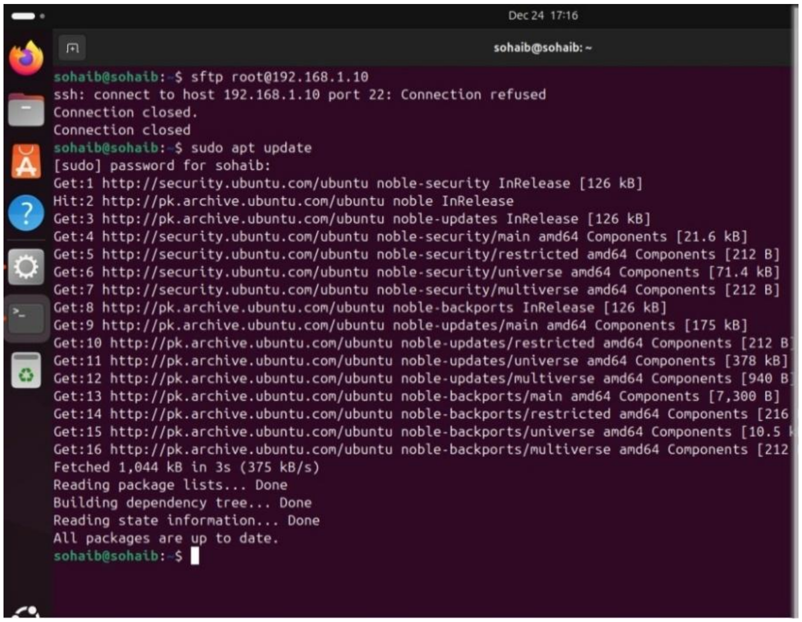
Command Name: sftp

Command Syntax: sftp [options] user@host

Description: Provides an interactive, encrypted file transfer session over SSH.

Dependencies/Installation: sudo apt install openssh-client

Screenshot:



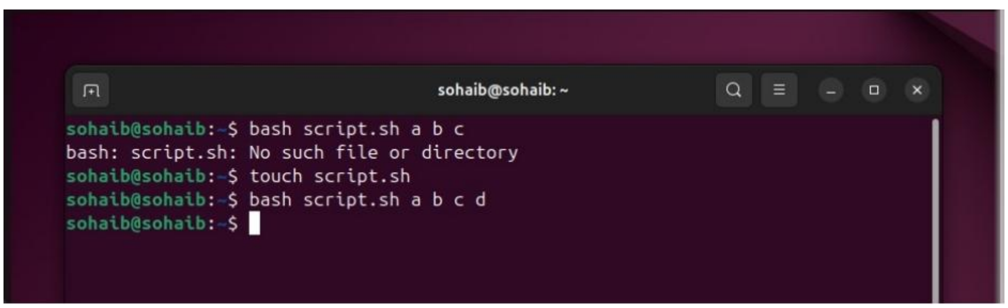
Command Name: shift

Command Syntax: shift [n]

Description: Shifts positional parameters to the left, discarding the first argument.

Dependencies/Installation: Built into the shell, no installation needed.

Screenshot:



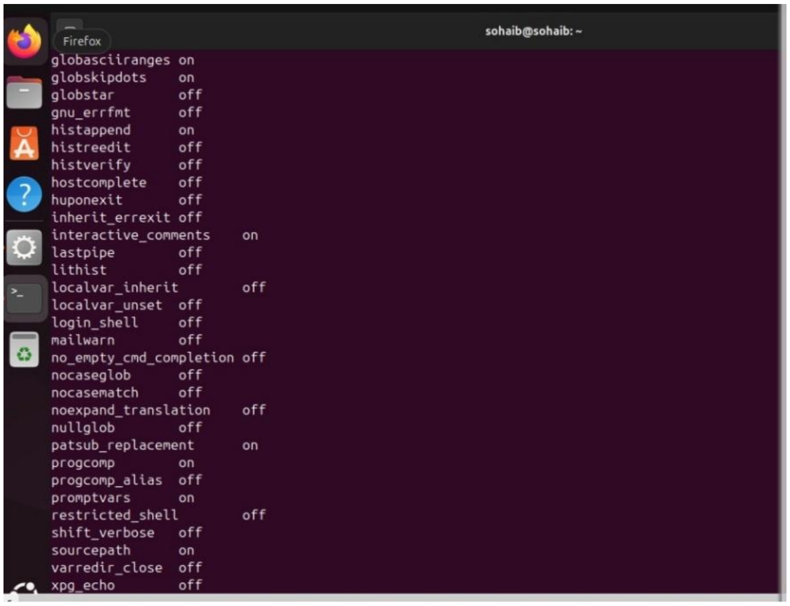
Command Name: shopt

Command Syntax: shopt [options] [optname ...]

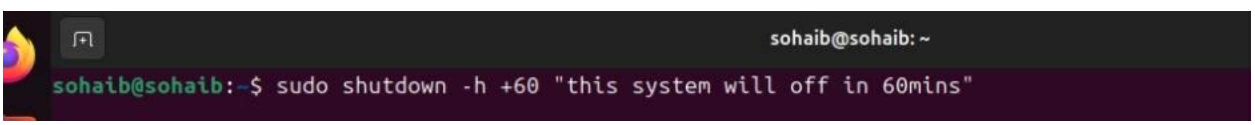
Description: Manages optional shell behavior settings in Bash.

Dependencies/Installation: Built into Bash, no installation needed.

Screenshot:



Command Name: shutdown
Command Syntax: sudo shutdown [options] time [message]
Description: Powers off, halts, or reboots the system at a specified time.
Dependencies/Installation: Typically pre-installed via system utilities.
Screenshot:



Command Name: sleep
Command Syntax: sleep number[suffix]
Description: Pauses execution for a specified number of seconds.
Dependencies/Installation: Part of coreutils, typically pre-installed.
Screenshot:

Command Name: slocate

Command Syntax: slocate [options] pattern

Description: Finds files by name using a pre-built, secure database.

Dependencies/Installation: sudo apt install mlocate

Screenshot:

Command Name: sort

Command Syntax: sort [options] [file]

Description: Sorts lines of text files alphabetically, numerically, or by other criteria.

Dependencies/Installation: Part of coreutils, typically pre-installed.

Screenshot:

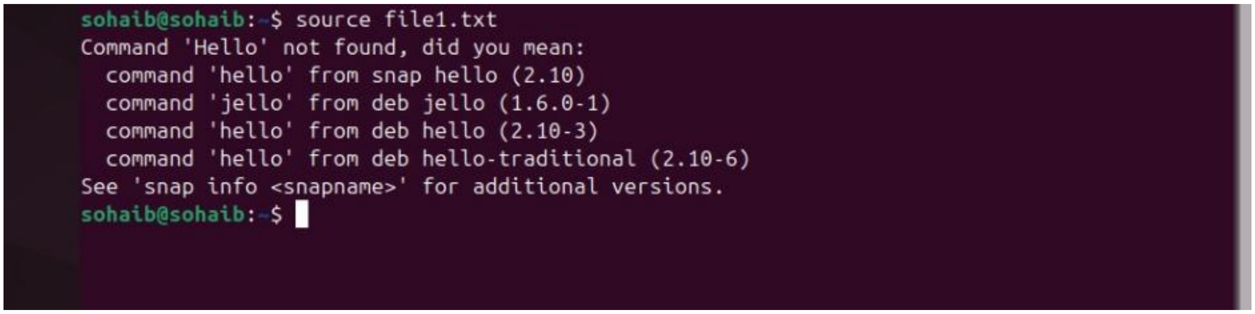
Command Name: source

Command Syntax: source filename or . filename

Description: Executes commands from a file in the current shell environment.

Dependencies/Installation: Built into the shell, no installation needed.

Screenshot:



Command Name: split

Command Syntax: split [options] input_file [prefix]

Description: Splits a large file into smaller pieces by lines or bytes.

Dependencies/Installation: Part of coreutils, typically pre-installed.

Screenshot:

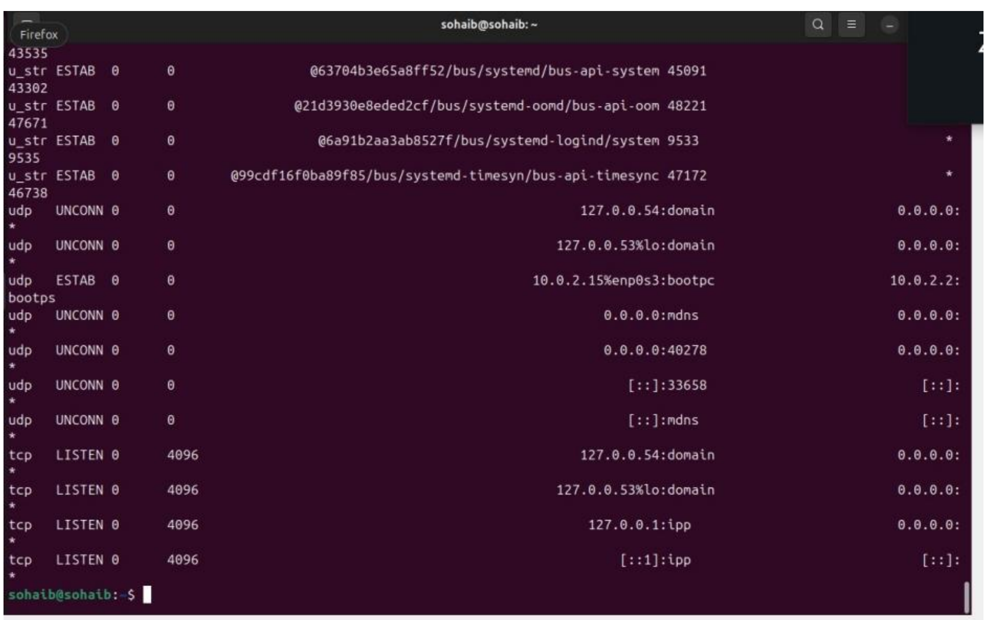
Command Name: ss

Command Syntax: ss [options] [filter]

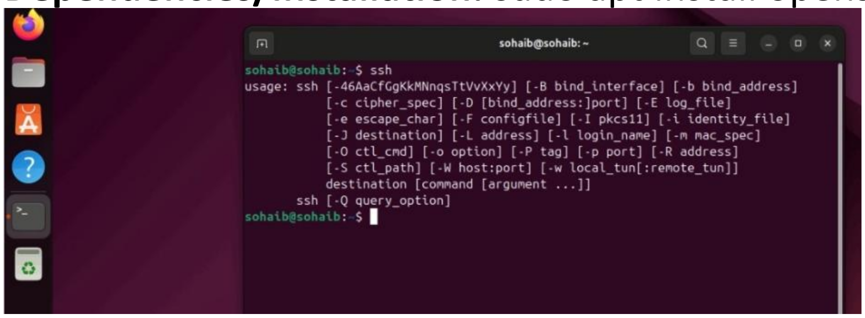
Description: Displays detailed information about network sockets.

Dependencies/Installation: Part of iproute2, typically pre-installed.

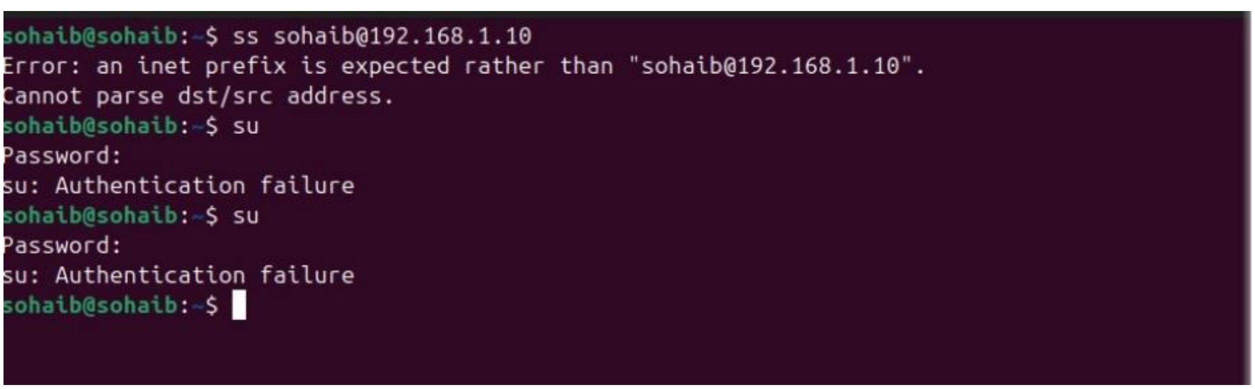
Screenshot:



Command Name: ssh
Command Syntax: ssh [options] user@host [command]
Description: Establishes an encrypted, secure remote login session.
Dependencies/Installation: sudo apt install openssh-client



Command Name: su
Command Syntax: su [options] [username]
Description: Switches the current user to another user within a login shell.
Dependencies/Installation: Part of util-linux, typically pre-installed.
Screenshot:



Command Name: sudo
Command Syntax: sudo [options] command
Description: Executes a command with superuser or another user's privileges.

Dependencies/Installation: sudo apt install sudo

Screenshot:

Command Name: sum

Command Syntax: sum [options] file

Description: Calculates a checksum and block count for a file (legacy command).

Dependencies/Installation: Part of coreutils, typically pre-installed.

Screenshot:

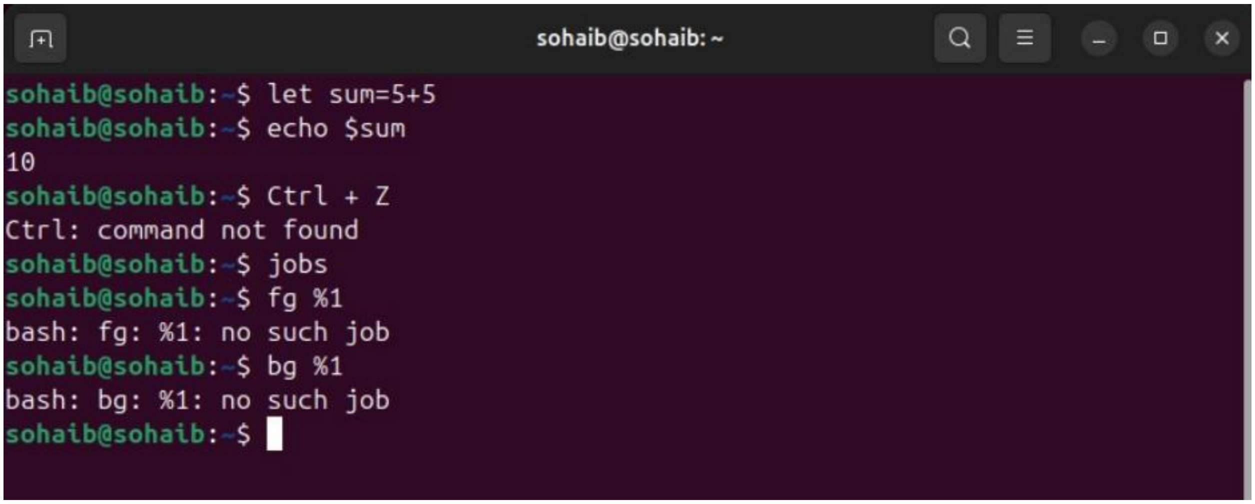
Command Name: suspend

Command Syntax: suspend [-f]

Description: Suspends the execution of the current shell.

Dependencies/Installation: Built into Bash, no installation needed.

Screenshot:



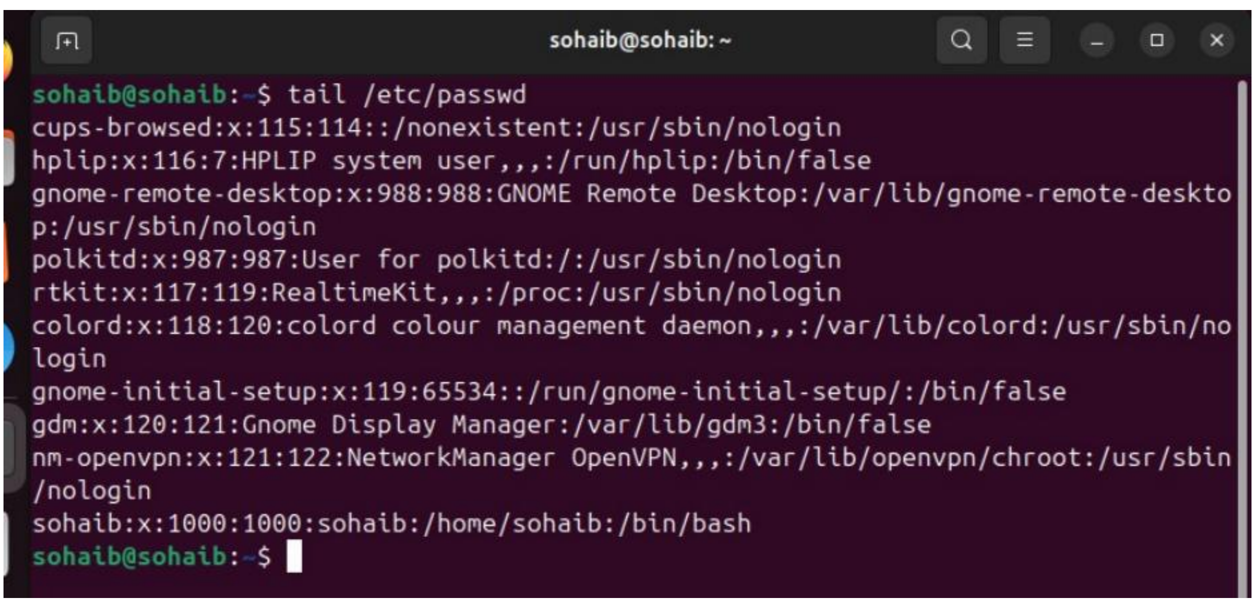
Command Name: tail

Command Syntax: tail [options] [file]

Description: Outputs the last part (default 10 lines) of files.

Dependencies/Installation: Part of coreutils, typically pre-installed.

Screenshot:



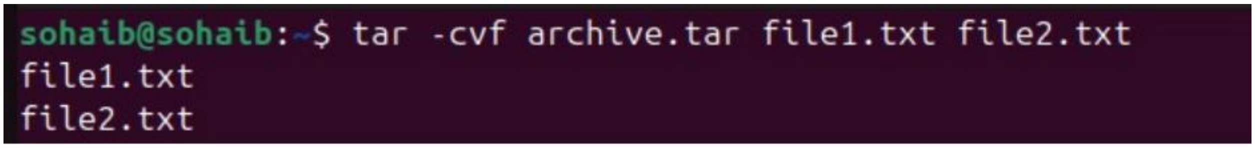
Command Name: tar

Command Syntax: tar [options] [archive_file] [files_to_archive]

Description: Creates, extracts, and manipulates archive files.

Dependencies/Installation: Part of tar package, typically pre-installed.

Screenshot:



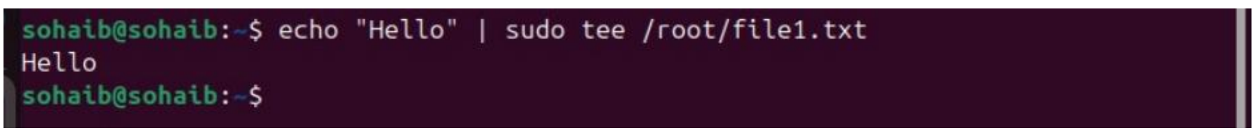
Command Name: tee

Command Syntax: command | tee [options] file

Description: Reads from stdin and writes to both stdout and one or more files.

Dependencies/Installation: Part of coreutils, typically pre-installed.

Screenshot:



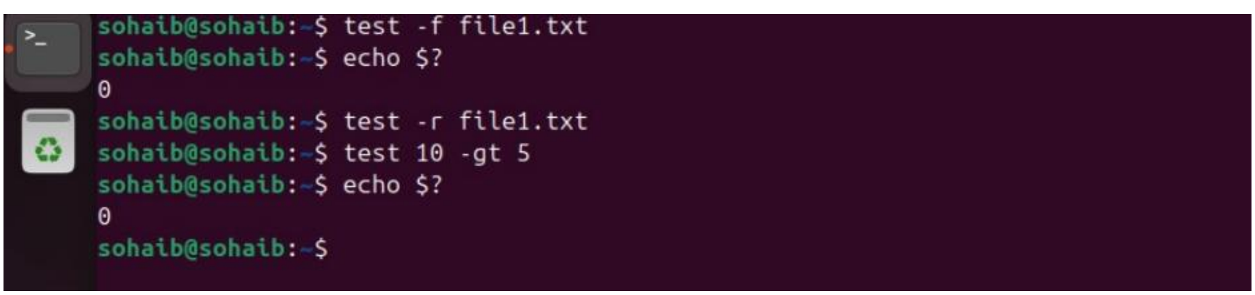
Command Name: test

Command Syntax: test expression or [expression]

Description: Evaluates conditional expressions and returns a status.

Dependencies/Installation: Built into the shell, no installation needed.

Screenshot:

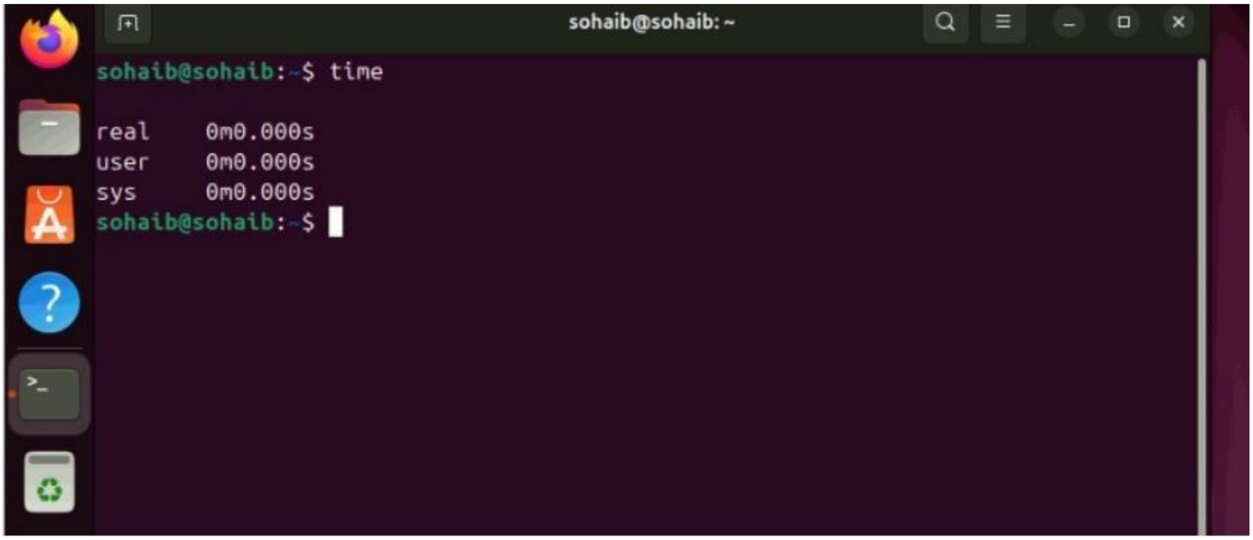


Command Name: time

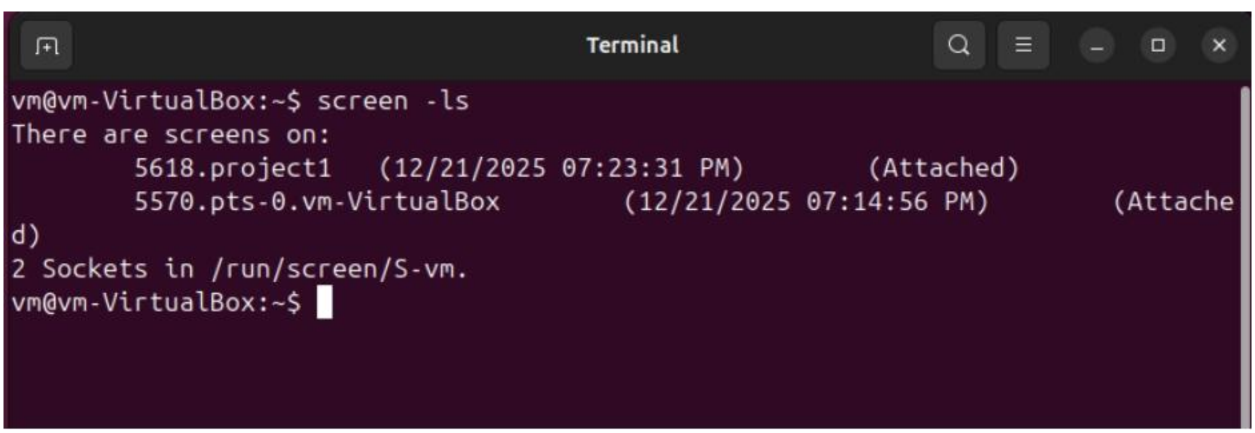
Command Syntax: time [options] command [arguments]

Description: Measures the execution time of a command.

Dependencies/Installation: Part of time package or shell built-in.
Screenshot:



Command Name: timeout
Command Syntax: timeout [options] duration command [arguments]
Description: Runs a command with a time limit, terminating it if exceeded.
Dependencies/Installation: Part of coreutils, typically pre-installed.
Screenshot:

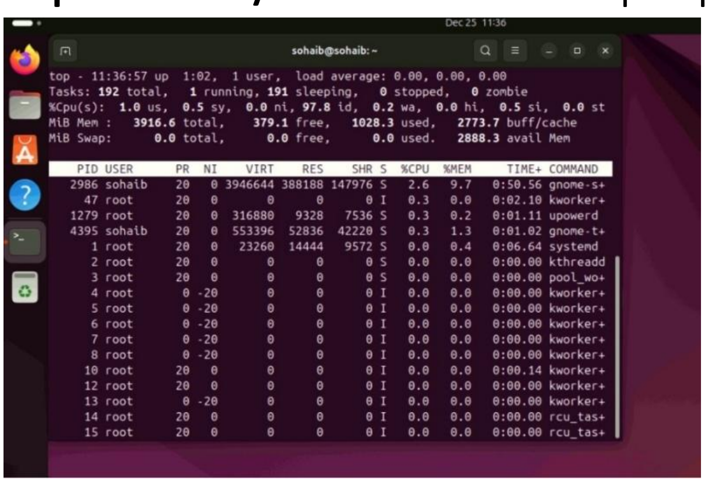


Command Name: times
Command Syntax: times

Description: Displays accumulated user and system CPU times for the shell.
Dependencies/Installation: Built into the shell, no installation needed.
Screenshot:

Command Name: touch
Command Syntax: touch [options] file
Description: Creates empty files or updates timestamps of existing files.
Dependencies/Installation: Part of coreutils, typically pre-installed.
Screenshot:

Command Name: top
Command Syntax: top [options]
Description: Displays a dynamic, real-time view of system processes.
Dependencies/Installation: Part of procps, typically pre-installed.



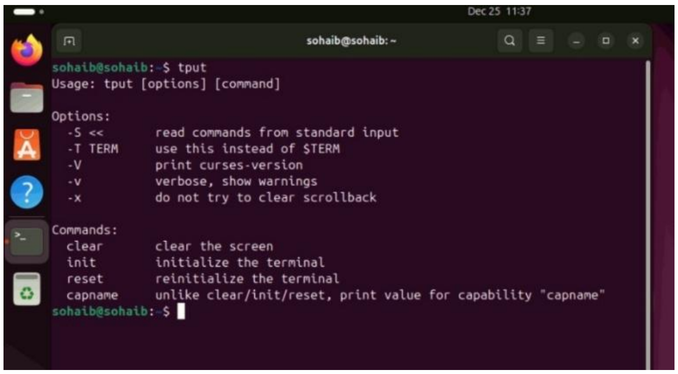
Command Name: tput

Command Syntax: tput [option] [capability_name]

Description: Queries and manipulates terminal capabilities.

Dependencies/Installation: Part of ncurses-bin, typically pre-installed.

Screenshot:



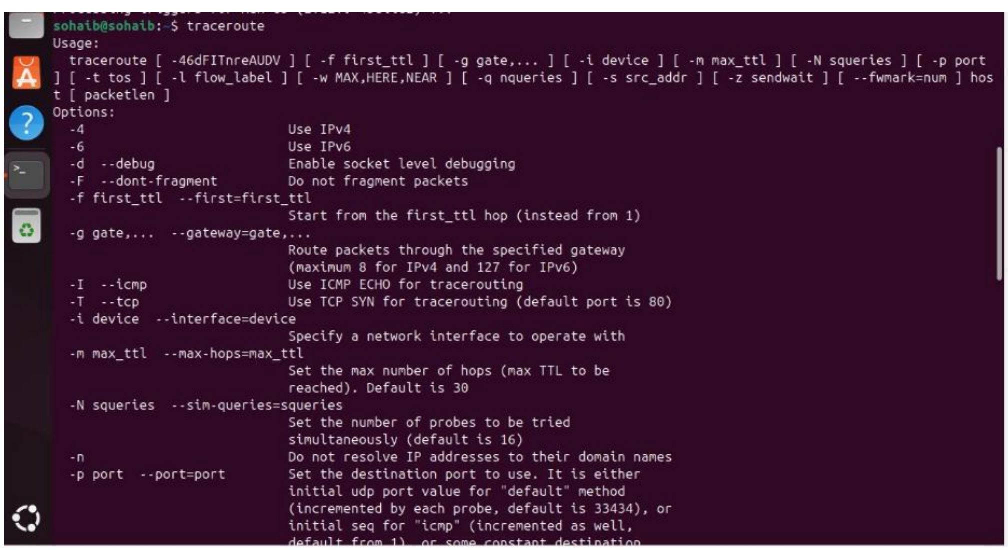
Command Name: traceroute

Command Syntax: traceroute [options] host

Description: Traces the network path packets take to a destination host.

Dependencies/Installation: sudo apt install traceroute

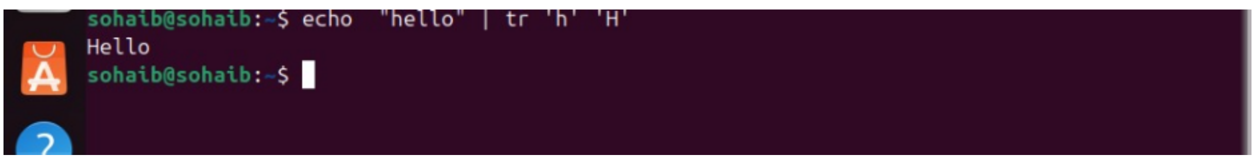
Screenshot:



Command Name: trap
Command Syntax: trap 'commands' signals
Description: Specifies commands to execute when the shell receives signals.
Dependencies/Installation: Built into the shell, no installation needed.
Screenshot:

The trap command may not work if it is used in the wrong shell, written with incorrect signal syntax, or the script exits before the signal is generated.

Command Name: tr
Command Syntax: tr [options] set1 set2
Description: Translates, squeezes, or deletes characters from input.
Dependencies/Installation: Part of coreutils, typically pre-installed.
Screenshot:



Command Name: true
Command Syntax: true
Description: Returns a successful (zero) exit status.
Dependencies/Installation: Built into the shell, no installation needed.
Screenshot:

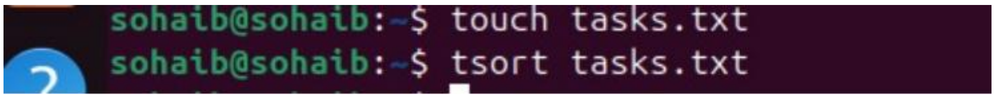
Command Name: tsort

Command Syntax: tsort [options] [file]

Description: Performs a topological sort on partially ordered items.

Dependencies/Installation: Part of coreutils, typically pre-installed.

Screenshot:



Command Name: type

Command Syntax: type [options] command_name

Description: Indicates how a command name is interpreted by the shell.

Dependencies/Installation: Built into the shell, no installation needed.

Screenshot:

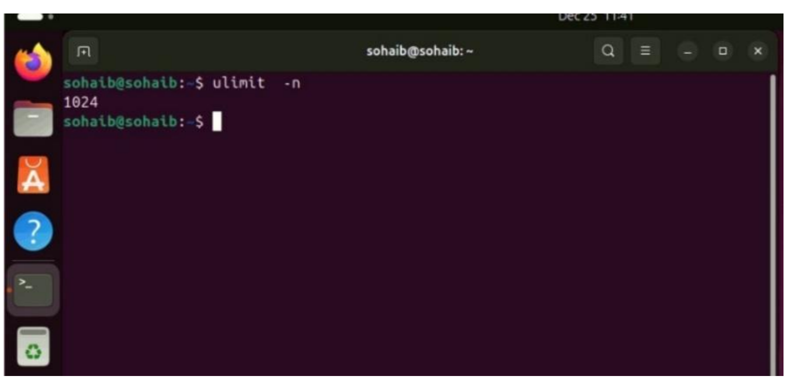
Command Name: ulimit

Command Syntax: ulimit [options] [limit]

Description: Gets or sets user-level limits on system resources.

Dependencies/Installation: Built into the shell, no installation needed.

Screenshot:



Command Name: umask
Command Syntax: umask [options] [mask]
Description: Sets or displays the file mode creation mask.
Dependencies/Installation: Built into the shell, no installation needed.
Screenshot:

Command Name: umount
Command Syntax: sudo umount [options] mount_point
Description: Detaches a previously mounted filesystem.
Dependencies/Installation: Part of util-linux, typically pre-installed.
Screenshot:

Command Name: unalias
Command Syntax: unalias [options] alias_name
Description: Removes alias definitions from the current shell.
Dependencies/Installation: Built into the shell, no installation needed.
Screenshot:

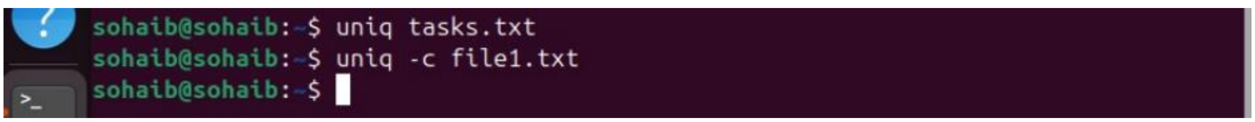
Command Name: uniq

Command Syntax: uniq [options] [input_file]

Description: Filters out or reports adjacent duplicate lines from sorted input.

Dependencies/Installation: Part of coreutils, typically pre-installed.

Screenshot:



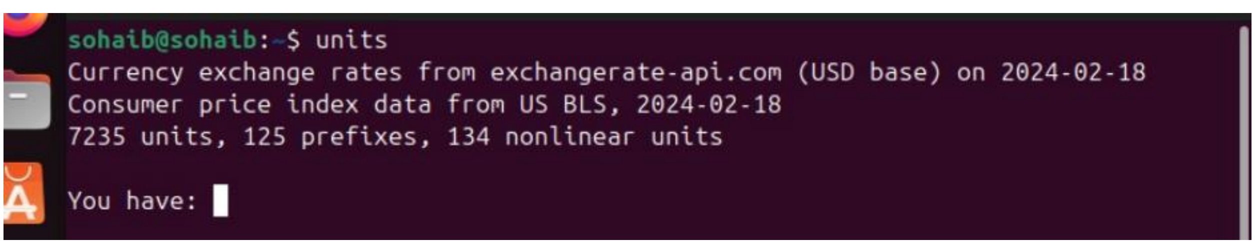
Command Name: units

Command Syntax: units [options] [from_unit to_unit]

Description: Interactive program for converting between units of measurement.

Dependencies/Installation: sudo apt install units

Screenshot:



Command Name: until

Command Syntax: until test-commands; do consequent-commands; done

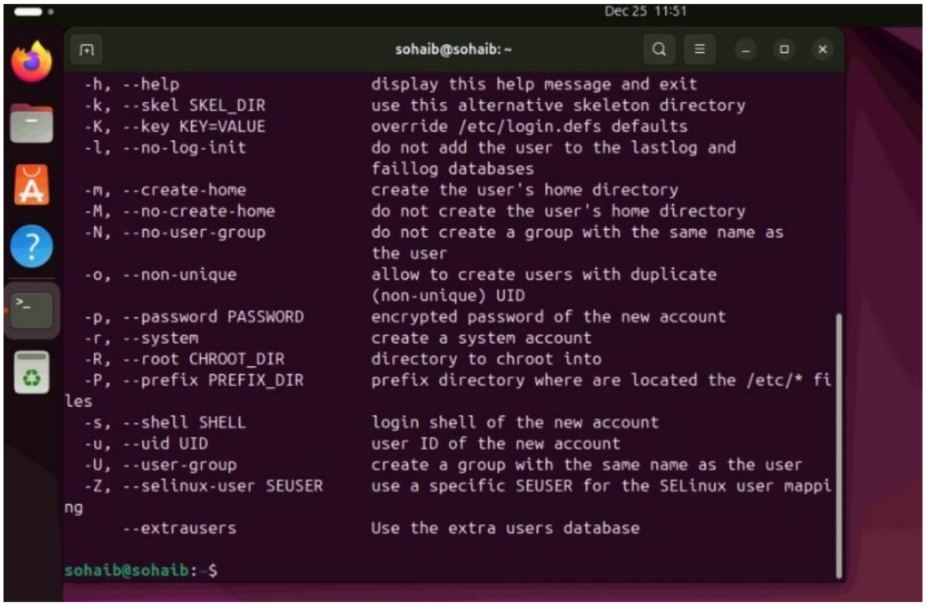
Description: Loop that executes commands until a condition becomes

true.
Dependencies/Installation: Built into the shell, no installation needed.
Screenshot:

The until loop does not run because its condition is already true at the start, so the loop terminates immediately.

Command Name: uptime
Command Syntax: uptime [options]
Description: Shows system uptime, current time, user count, and load averages.
Dependencies/Installation: Part of procps, typically pre-installed.
Screenshot:

Command Name: useradd
Command Syntax: sudo useradd [options] username
Description: Creates a new user account or updates default new user info.
Dependencies/Installation: Part of passwd package, typically pre-installed.
Screenshot:



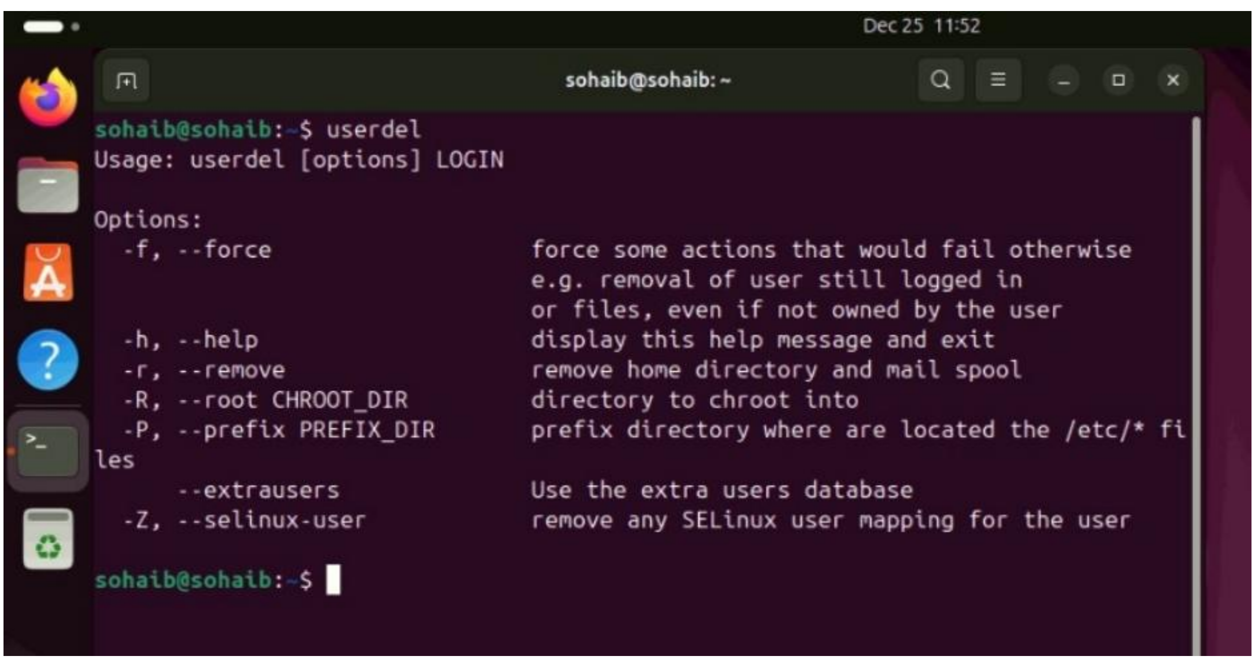
Command Name: userdel

Command Syntax: sudo userdel [options] username

Description: Deletes a user account and related files.

Dependencies/Installation: Part of passwd package, typically pre-installed.

Screenshot:

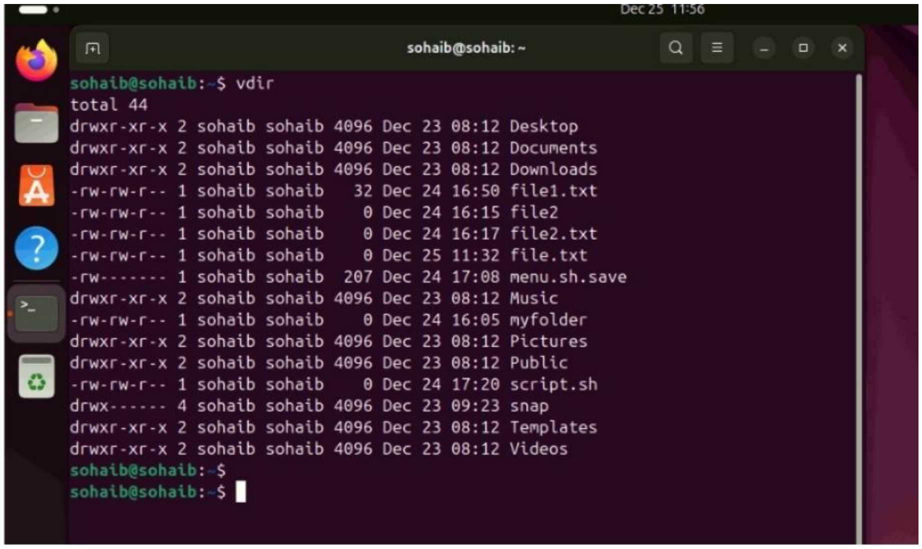


Command Name: usermod
Command Syntax: sudo usermod [options] username
Description: Modifies an existing user account's attributes.
Dependencies/Installation: Part of passwd package, typically pre-installed.
Screenshot:

User mode has limited privileges, so certain commands or operations cannot run due to lack of administrative (root) permissions.

Command Name: users
Command Syntax: users [options] [file]
Description: Lists usernames of users currently logged in.
Dependencies/Installation: Part of coreutils, typically pre-installed.
Screenshot:

Command Name: vdir
Command Syntax: vdir [options] [file]
Description: Lists directory contents in long format (equivalent to ls -l -b).
Dependencies/Installation: Part of coreutils, typically pre-installed.
Screenshot:



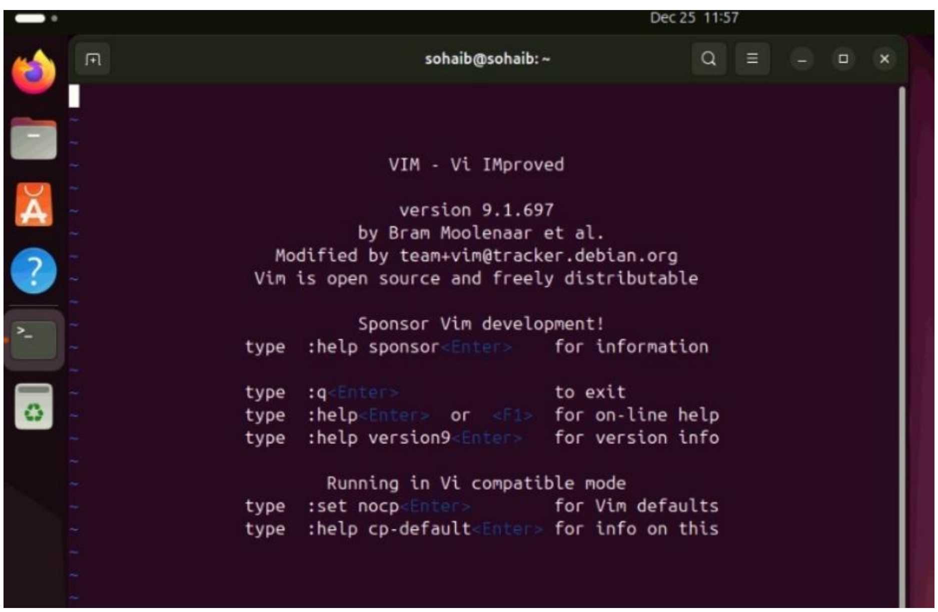
Command Name: vi

Command Syntax: vi [options] [file]

Description: A classic, screen-oriented text editor.

Dependencies/Installation: sudo apt install vim

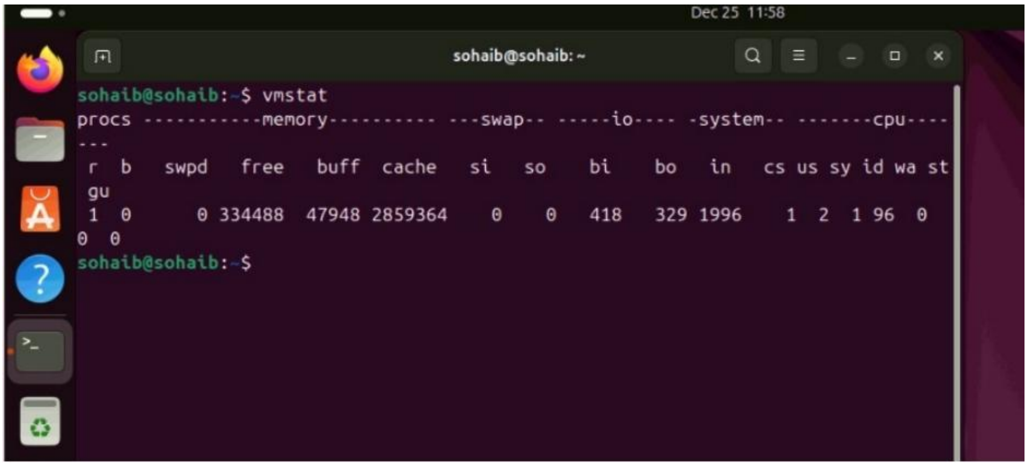
Screenshot:



Command Name: vmstat

Command Syntax: vmstat [options] [delay [count]]

Description: Reports virtual memory statistics and system activity.
Dependencies/Installation: Part of procps, typically pre-installed.
Screenshot:



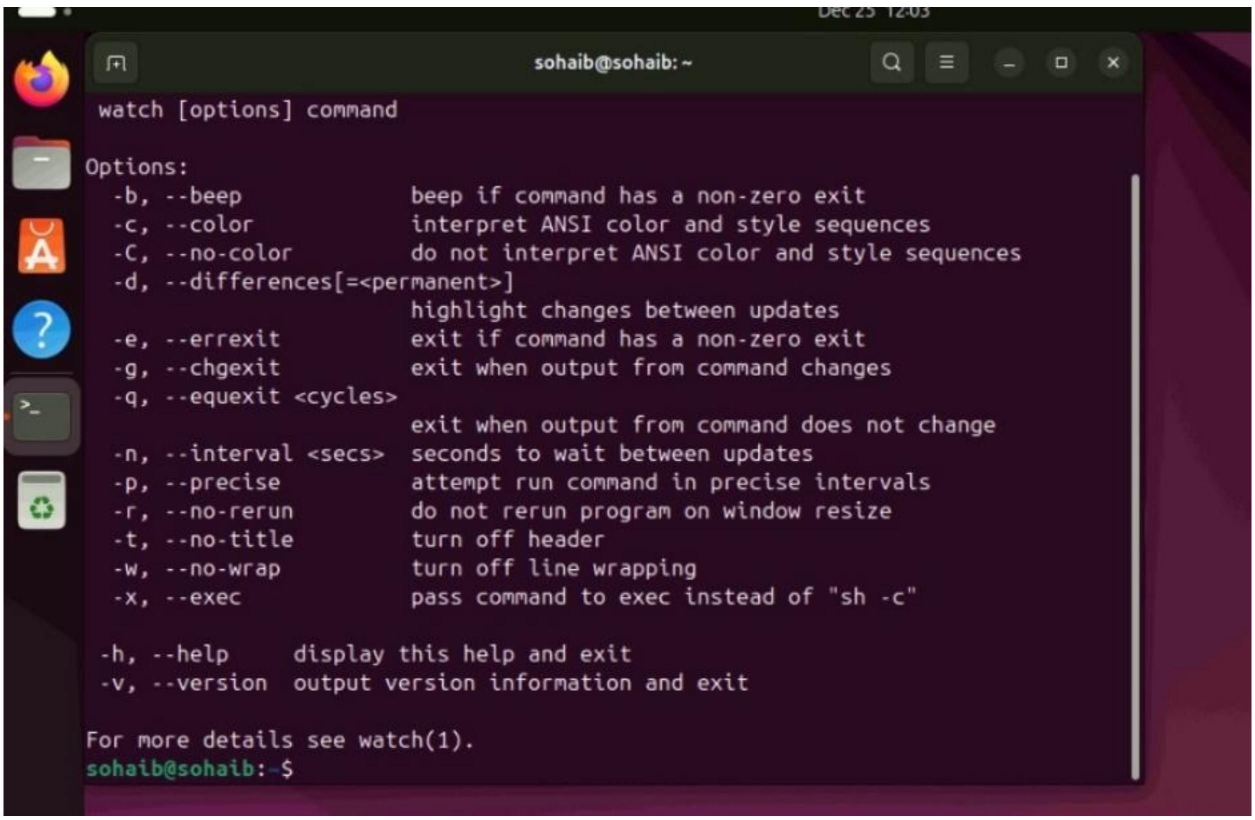
Command Name: w
Command Syntax: w [options] [user]
Description: Shows logged-in users and their processes, plus system load.
Dependencies/Installation: Part of procps, typically pre-installed.
Screenshot:

Command Name: wait
Command Syntax: wait [options] [pid ...]
Description: Waits for child processes to complete and returns their status.

Dependencies/Installation: Built into the shell, no installation needed.
Screenshot:

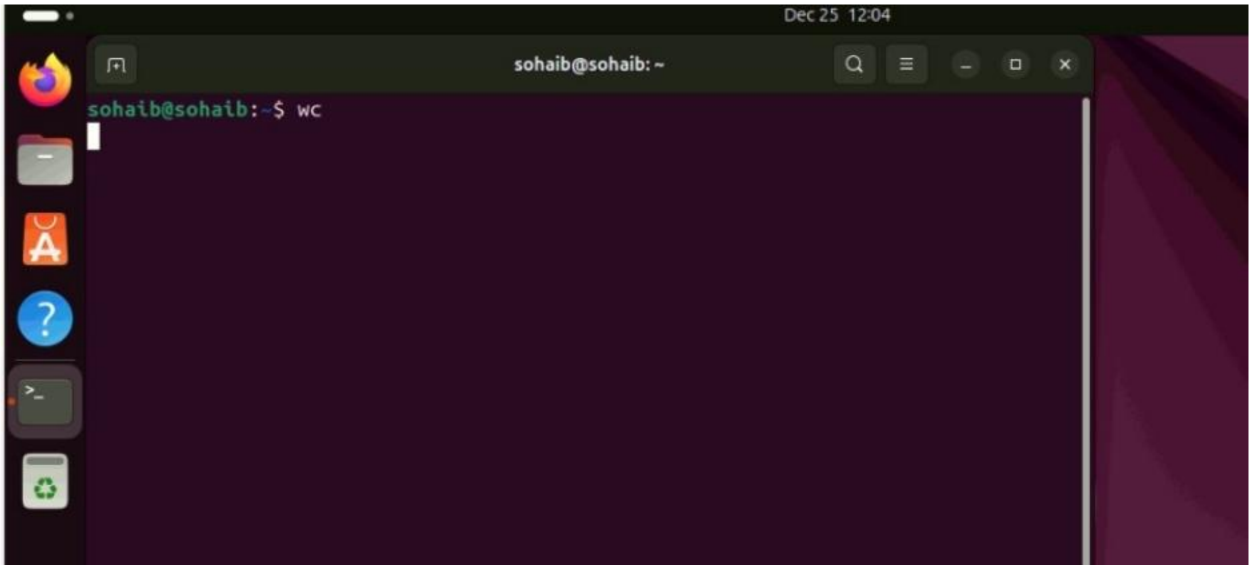
The wait command does not produce output unless it is waiting for a background process, so it appears not to run.

Command Name: watch
Command Syntax: watch [options] command
Description: Executes a command repeatedly at regular intervals.
Dependencies/Installation: sudo apt install watch
Screenshot:



Command Name: wc
Command Syntax: wc [options] [file]
Description: Counts lines, words, and bytes/characters in files.

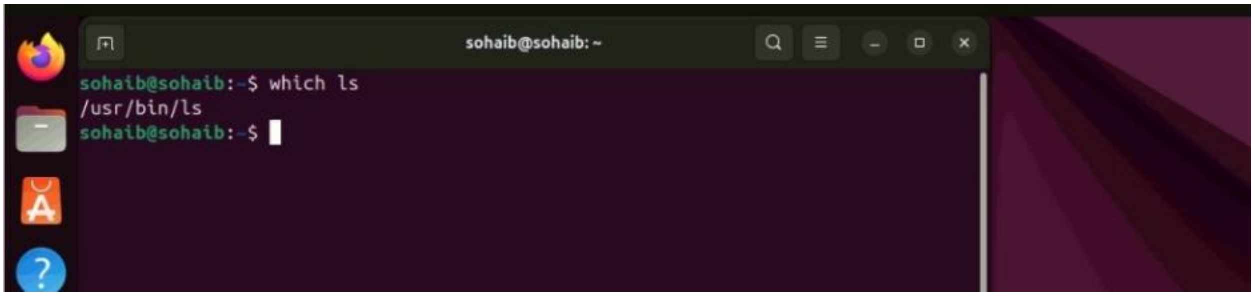
Dependencies/Installation: Part of coreutils, typically pre-installed.
Screenshot:



Command Name: whereis
Command Syntax: whereis [options] command_name
Description: Locates binary, source, and manual page files for a command.
Dependencies/Installation: Part of util-linux, typically pre-installed.
Screenshot:

The whereis command may show no output if the program is not installed or not present in standard binary directories.

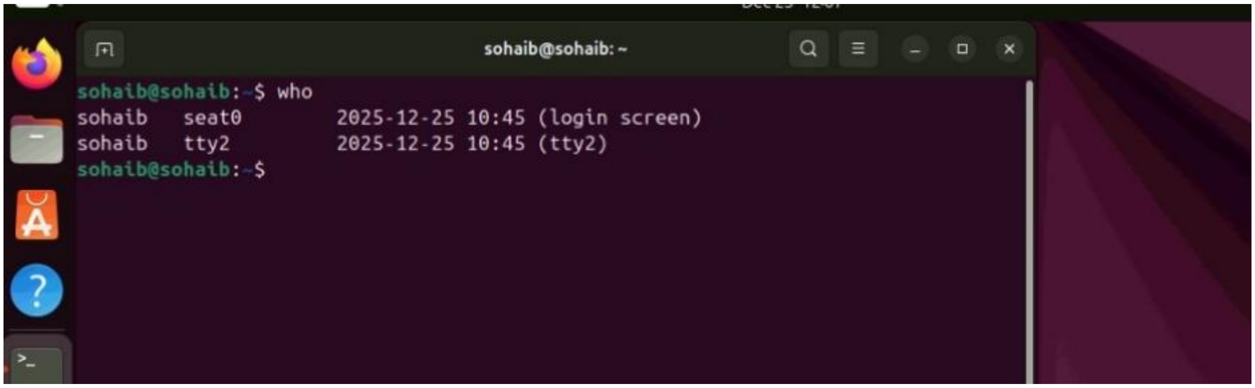
Command Name: which
Command Syntax: which [options] [--] command_name
Description: Displays the full path of executable commands.
Dependencies/Installation: Part of debianutils, typically pre-installed.
Screenshot:



Command Name: while
Command Syntax: while test-commands; do consequent-commands; done
Description: Loop that executes commands as long as a condition is true.
Dependencies/Installation: Built into the shell, no installation needed.
Screenshot:

The while loop does not execute because its condition is false at the beginning.

Command Name: who
Command Syntax: who [options] [file]
Description: Lists users currently logged into the system.
Dependencies/Installation: Part of coreutils, typically pre-installed.
Screenshot:



Command Name: whoami
Command Syntax: whoami
Description: Prints the effective username of the current user.
Dependencies/Installation: Part of coreutils, typically pre-installed.
Screenshot:

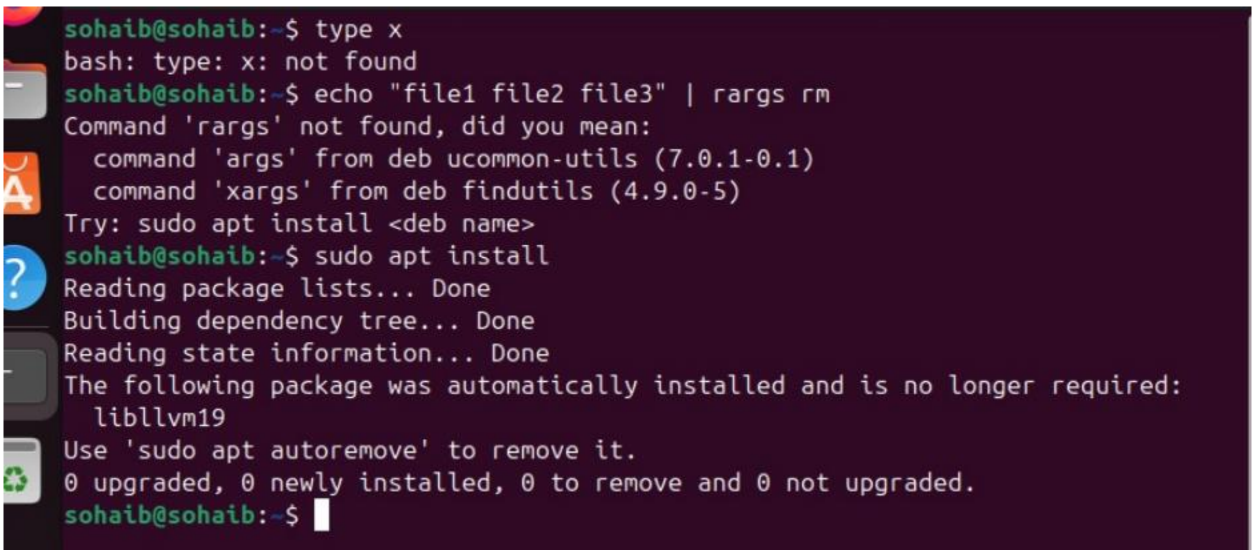
Command Name: wget
Command Syntax: wget [options] [URL]
Description: A non-interactive network downloader for HTTP, HTTPS, and FTP.
Dependencies/Installation: sudo apt install wget
Screenshot:

wget may fail if it is **not installed**, the **URL is incorrect**, there is **no internet connection**, or the system **blocks network access**.

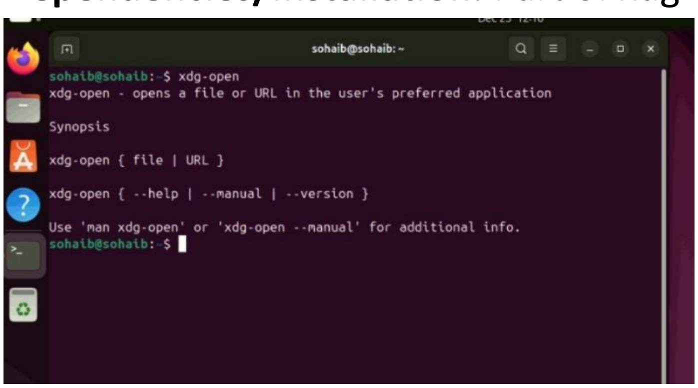
Command Name: write
Command Syntax: write user [terminal]
Description: Sends a message to another logged-in user on the same system.
Dependencies/Installation: Part of util-linux, typically pre-installed.
Screenshot:

The target user is not logged in or has disabled message receiving (mesg n).

Command Name: xargs
Command Syntax: command | xargs [options] [command]
Description: Builds and executes command lines from standard input.
Dependencies/Installation: Part of findutils, typically pre-installed.
Screenshot:



Command Name: xdg-open
Command Syntax: xdg-open {file|URL}
Description: Opens a file or URL using the user's preferred application.
Dependencies/Installation: Part of xdg-utils, typically pre-installed.



Command Name: xz
Command Syntax: xz [options] [file]
Description: Compresses or decompresses files using LZMA/LZMA2 algorithm.
Dependencies/Installation: sudo apt install xz-utils
Screenshot:

Command Name: yes
Command Syntax: yes [string]
Description: Outputs a string repeatedly until killed (default "y").
Dependencies/Installation: Part of coreutils, typically pre-installed.
Screenshot:

Command Name: zip
Command Syntax: zip [options] zipfile list_of_files
Description: Packages and compresses files into a .zip archive.
Dependencies/Installation: sudo apt install zip
Screenshot:

The zip command may not produce output if the **file/directory does not exist**, the **syntax is wrong**, or the **zip package is not installed**.